

The Self-Auditing Ledger: Leakage-Free Temporal Graph Topology for ERP Fraud Detection, with a Production Audit Console

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Abstract—Graph-based fraud detection methods are frequently proposed without an explicit audit of temporal leakage, and rarely evaluated with a controlled, same-architecture ablation that isolates the marginal value of graph structure from the tabular signal it is layered on top of. This paper presents a leakage-free streaming temporal-topology feature engine – enforcing a formal read-then-insert invariant so that a transaction’s features can never depend on information from its own future – and subjects it to a rigorous within-dataset ablation across four heterogeneous financial datasets (a bipartite payment simulator, a multi-hop transfer network, a real SAP general-ledger export, and a calibrated synthetic ERP economy). The central finding is conditional, not universal: graph topology adds substantial, leakage-free predictive value on two of four datasets (+11.7% and +246.8% relative PR-AUC), is uninformative on a third whose tabular baseline is already saturated, and *actively hurts* on the fourth. We report this honestly, including the finding that hand-crafted cycle-detection features – the project’s own headline motif – carry near-zero gain importance on three of four datasets, that a learned temporal graph network never beats hand-crafted topology on real data, and that naive synthetic-data augmentation measurably harms real-dataset performance. Beyond the research question, we describe and empirically verify a production system – a Procure-to-Pay ERP with a forensic audit console – that serves this leakage-free feature computation live against real transactions, with database-enforced financial-integrity invariants (double-entry balance, period close, immutable audit trail, and a five-layer duplicate-payment defence proven under concurrent load), closing the gap between a research pipeline and a system an auditor could actually use.

Index Terms—fraud detection, temporal graphs, data leakage, anti-money laundering, explainable AI, continuous auditing, ERP systems, XGBoost, synthetic data

I. INTRODUCTION

Enterprise resource planning (ERP) systems are the ledger of record for most organized economic activity: procurement, payroll, intercompany transfer, and customer settlement all resolve, eventually, into a stream of timestamped postings between accounts. Financial fraud and money laundering exploit this same stream, but rarely as an isolated anomalous posting. The characteristic shapes of laundering and payment fraud – a receiving account funded by dozens of unrelated senders, a chain of transfers that closes on itself days later, a set of near-identical amounts recurring between two parties

– are properties of *several transactions taken together*, not of any single row in a table. A model that scores each transaction independently, however strong its tabular features, is structurally blind to this class of evidence: two transactions with identical amount, timestamp, and account features can carry completely different fraud risk depending on what else has moved through the same accounts. This is the motivating premise for representing ERP data as a graph – accounts as nodes, transactions as time-stamped edges – and for asking whether the graph’s *topology* carries predictive signal that a transaction-level baseline does not already capture.

That question is easy to ask and, in the graph-based fraud detection literature, surprisingly rarely answered cleanly. Two problems recur. First, temporal graph features are leakage-prone in a way ordinary tabular features are not: a naively implemented graph statistic (an account’s total degree, a cycle-membership flag, a rolling average) computed over a graph built from the *whole* dataset will, by construction, let a transaction’s features depend on transactions that occur after it in time – the graph equivalent of letting a model see tomorrow’s newspaper. Second, when a paper reports that adding graph structure improves a fraud metric, the improvement is frequently measured against a baseline of unstated or weaker capacity, so the reader cannot tell whether the gain reflects genuine structural signal or simply a better-tuned model. This project’s own history is a cautionary instance of both failures at once. An earlier iteration of the pipeline described in this paper claimed graph topology as its central differentiator; on audit, its own feature-importance output showed every topology feature scoring effectively zero gain importance, while a top-ranked feature turned out to be the time until the *next* transaction – future information leaking directly into a past transaction’s row. Its headline accuracy numbers were consequently uninterpretable. Rebuilding the pipeline so that this failure mode is structurally impossible, and then running the first fair test of the original question, is the starting point of this paper.

A. The Scientific Claim

We deliberately narrow the question this paper answers. It would be easy to frame the contribution as a real-time fraud-monitoring platform for ERP systems – and the system

sections later in this paper do describe a working console that ingests postings, computes features online, and surfaces explained alerts. But a working system is a demonstration, not evidence of the platform’s founding hypothesis, and this paper treats it accordingly: as the *vision* the architecture is built toward, not the *proof* it is asked to supply. The proof this paper commits to is narrower, falsifiable, and stated once, precisely, here:

Temporal graph topology features add predictive value over a strong tabular baseline, measured without leakage.

Every other claim in this paper – about the streaming architecture, the synthetic data generator, the production console, the explanation layer – is in service of being able to answer that question honestly, on a real dataset, without the answer being pre-determined by how the experiment was built. We committed, before running the decisive experiment, to reporting whatever it showed, including a negative result on any of the four datasets tested. As quantified later in this paper, the honest answer turns out to be conditional: topology helps substantially where the tabular baseline has headroom and fraud leaves a structural trace, is uninformative where the baseline is already near-saturated by amount alone, and on the one dataset with genuinely thin ERP-style graph structure and very few labeled test frauds, topology measurably *hurts* the full model. We report that last result plainly rather than omitting it, because a paper that sets out to test whether topology helps is not entitled to discard the trial where it did not.

B. Leakage as a Design Constraint, Not a Post-hoc Check

The central engineering response to the leakage failure above is a streaming computation model applied uniformly to every topology feature, rather than a leakage *test* applied after the fact to features computed some other way. Transactions are processed in a single pass, ordered by timestamp with ties broken by a fixed, arbitrary total order over the input rows. Let $\tau_1, \tau_2, \dots, \tau_n$ denote this processing order and let G_i denote the graph state after transactions $\tau_1, \dots, \tau_i$ have been inserted, with $G_0 = \emptyset$. For every transaction τ_i , the engine enforces

$$\phi(\tau_i) = f(G_{i-1}), \quad \text{then} \quad G_i \leftarrow G_{i-1} \cup \{\tau_i\},$$

i.e., the feature vector $\phi(\tau_i)$ is computed and persisted against the graph state that excludes τ_i itself and everything after it, and only *then* is τ_i inserted. We refer to this as the *read-then-insert invariant*. It is not merely a coding convention: a structural property that once a feature vector is written it is provably a function of τ_i ’s strict causal past means a later transaction can be proven, not merely assumed, to be unable to alter an earlier transaction’s features. A later section of this paper formalizes the leakage taxonomy this invariant closes off, and the accompanying automated regression test – adversarially scrambling the order of not-yet-processed transactions and asserting that no already-computed feature changes – has stayed green through every stage of this project’s rebuild. Figure 1 sketches the resulting pipeline end to end, including the point at which the deployed system diverges

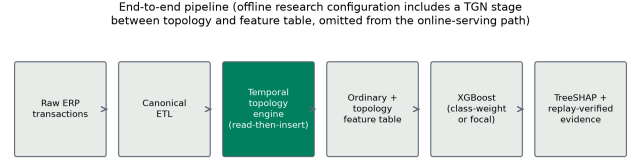


Fig. 1: End-to-end pipeline. Raw ERP transaction feeds are cleaned into a canonical accounts/transactions schema, assembled into a temporal shadow graph, and passed through the streaming topology engine under the read-then-insert invariant, so that a feature is provably a function only of transactions strictly earlier in processing order. The resulting feature table (ordinary + topology, with offline-only TGN embeddings added for the research configuration) feeds a gradient-boosted model; the console layer attaches a TreeSHAP explanation and a re-derived evidence path to every alert. The production/online bundle omits the TGN component, since its learned memory tensor is not persisted for live inference.

from the research configuration used to establish the scientific claim above, elaborated fully in the system description later in this paper.

C. Contributions

This paper makes five contributions, listed in the order the paper substantiates them:

- 1) **A leakage-free streaming topology engine with a formal read-then-insert invariant.** Every one of the engine’s structural detectors (in-/out-degree fan-in and fan-out, cycle membership and amount-conserving cycle features, and recurring near-equal-amount “lapping” features) is computed under the invariant above and guarded by an automated, order-scrambling regression test, rather than validated only by an after-the-fact audit.
- 2) **A rigorous within-dataset ablation across four heterogeneous datasets.** The same feature engine, model family, and evaluation protocol are applied, unchanged, to two independent real-world payment datasets, one real ERP general-ledger dataset, and one internally generated synthetic dataset chosen to span a deliberate spectrum of graph structure, from near-tree (customer-to-merchant only, cycles structurally impossible) to rich, cyclical, multi-account flows. Reporting the same probe on datasets that differ this much in how much structure they even contain is itself part of the evidence: a topology signal that appears only where structure is rich and vanishes where it is sparse is more credible than one reported on a single dataset.
- 3) **A calibrated synthetic ERP economy generator with an adversarial realism audit.** To study topology-rich fraud behavior and to probe whether synthetic data can safely augment real training sets, we build a seeded, agent-based synthetic ERP economy whose fraud is generated from criminal goals and evasion behavior rather than from the shapes our own detectors look for –

avoiding the circularity of grading a detector against data manufactured to fit it – and subject it to an independent adversarial review and a realism battery before using it for anything beyond sanity checks.

- 4) **A production system that serves the leakage-free feature computation online.** The same streaming engine that produces the research feature table also backs a running ERP application and forensic audit console, computing ordinary and topology features on live postings under the identical read-then-insert invariant. We are explicit throughout this paper about a consequential asymmetry: the strongest *offline* research configuration additionally includes learned temporal-graph-network (TGN) embeddings, but the TGN’s per-account memory state is not persisted, is not guaranteed bit-reproducible across machines even under a fixed random seed (quantified later in this paper), and is therefore never part of what is actually served online. Every PR-AUC figure in this paper is labeled as belonging to one of two distinct models – the *offline champion* (topology and, where used, TGN) or the *deployed online bundle* (ordinary and topology only) – and the two are never conflated.
- 5) **Full transparency, including negative and null results.** Alongside the positive result, we report, without softening: a dataset on which the full model’s topology addition measurably reduces PR-AUC; a dataset whose tabular baseline is already so strong that the marginal comparison is close to uninformative; an ablation study that is single-seed and therefore silent on whether a small positive lift is signal or noise; a headline structural-motif family (cycle detection) that contributes negligible gain importance on most of the datasets studied despite being the project’s original motivating idea; and a synthetic-data augmentation experiment whose main finding is that naive augmentation *degrades* real-data performance, with only a narrower, different transfer claim holding up.

D. Preview of Findings

Across the two real payment-network datasets and the real ERP ledger dataset evaluated in the final, multi-seed model comparison, the picture is genuinely mixed, not uniformly positive. On the sparse, customer-to-merchant payment dataset – structurally incapable of containing true multi-hop cycles – adding hand-crafted topology features raises test PR-AUC from a 0.7870 ± 0.0700 ordinary-feature baseline to 0.9319 ± 0.0017 , and the fully tuned champion configuration reaches 0.9364 ± 0.0015 with perfect precision at the top 100 alerts. On the anti-money-laundering simulation dataset, whose ordinary-feature baseline already reaches 0.9920 ± 0.0026 PR-AUC, adding topology reaches a ceiling of 1.0000 ± 0.0000 – a result we flag, not celebrate, as an artifact of an already-saturated baseline rather than evidence of a large structural effect; the informative signal on this dataset lives in an amount-blind probe reported later, not in this marginal comparison. On the real, university-published SAP general-ledger dataset, the honest result runs the other way: the ordinary-feature baseline

reaches 0.1581 ± 0.0129 PR-AUC, and adding topology *lowers* it to 0.0717 ± 0.0095 ; only after a full hyperparameter search and a change of imbalance mechanism does the final champion claw back to 0.1557 ± 0.0717 , statistically indistinguishable from the ordinary baseline it started from. This dataset carries only 25 labeled test frauds, and we treat the result as under-powered rather than disproving the hypothesis outright – but we do not discard it, and we do not let the two datasets where topology clearly helps stand in for a result that, on this one, it measurably did not. We return to all three results with full detail, confidence caveats, and feature-level attribution later in this paper.

E. Paper Organization

The remainder of this paper is organized into four further sections. Section II situates this paper in the graph-based fraud detection, class-imbalance, temporal graph network, continuous-auditing, and explainability literatures, and argues that this paper’s central gap-filling contribution is methodological – an explicit leakage taxonomy and a same-architecture, multi-dataset ablation – rather than a new graph architecture. Section III (Proposed Model and Methodology) develops the leakage taxonomy and its automated guard in full, describes the four evaluation datasets and the streaming topology engine, specifies the shared modeling and evaluation protocol, details the design of the synthetic-economy generator and the two-layer explanation mechanism, and closes with the production system’s architecture and database-enforced financial-integrity invariants. Section IV (Results and Analysis) reports every empirical outcome those methods produce – the topology feature-importance result, the seven-arm ablation across four datasets with full confidence caveats, the focal-loss and temporal-graph-network modernization study, the synthetic economy’s realism and augmentation results, the explainable-alert precision figures, the production system’s measured online/offline gap and load-test performance – and a consolidated discussion of threats to validity and limitations. Section V closes with the claim this paper actually supports, the production-engineering contribution independent of that claim, and three specifically scoped directions for future work.

II. RELATED WORK

This section reviews five bodies of work this paper draws on – graph-based fraud and money-laundering detection, class-imbalance learning, temporal graph networks, continuous auditing and ERP-centric fraud research, and post-hoc explainability – and closes by positioning this paper’s contribution against the first of these in particular.

A. Graph-Based Fraud and Anti-Money-Laundering Detection

The premise that fraud and laundering are relational, not point, phenomena has produced a substantial and fast-growing graph neural network (GNN) literature, surveyed broadly by Yuan et al. [1] for GNN-based anomaly detection and specifically for financial fraud by Motie and Raahemi [2], whose systematic review taxonomizes architectures, node/edge representations, and evaluation practices across the field. Within that

TABLE I: Representative recent graph-based approaches to financial fraud and anti-money-laundering detection cited in this paper.

Source	Method family	Stated focus
[2]	Systematic review	GNN architectures for financial fraud, taxonomized
[1]	Survey	GNN-based anomaly detection broadly, fraud as one application
[8]	Quantum GNN	Fraud classification via quantum graph circuits
[9]	Quantum-classical hybrid	Fraud detection with quantum-assisted feature selection
[3]	RL + GNN fusion	Real-time fraud scoring via RL-guided graph policies
[4]	GNN + imbalance learning	Transaction-level fraud detection under class skew
[5]	Conditional GAN + embeddings	Synthetic minority generation for imbalanced fraud data
[6]	Balanced GNN architecture	Architecture-level remedy for imbalanced graph classification
[7]	Contrastive uncertainty-aware GNN	+ Spatio-temporal fraud detection with calibrated confidence
[10]	RL causal explainer	Post-hoc causal explanation for GNN predictions

space, several recent directions are represented in the citations this paper draws on. Devi et al. [3] fuse reinforcement learning with a GNN backbone for real-time fraud detection, aiming at the same online-scoring setting this paper’s production console targets. Tong et al. [4] and Zhi et al. [5] both target the fraud literature’s dominant practical obstacle – extreme class imbalance – from the graph side, combining GNN representations with imbalance-aware learning and, respectively, an embedding-aware conditional GAN for synthetic minority generation. Boyapati et al. [6] propose an architecture-level balancing mechanism (BalancerGNN) rather than a data-level one. Shi et al. [7] combine spatio-temporal contrastive representation learning with explicit uncertainty quantification, addressing a concern this paper shares – that a bare point-estimate fraud score is of limited use to an auditor who must decide how much to trust it. A smaller quantum-computing strand is also represented: Innan et al. [8] and Grossi et al. [9] both explore quantum and hybrid quantum-classical graph learning for fraud classification and feature selection; both are, by the nature of current quantum hardware, proof-of-concept studies rather than deployment-scale systems, and we include them here for completeness of the graph-fraud landscape rather than as a direct comparison point for the tabular-scale production system this paper describes. Table I summarizes the method family and stated focus of each.

What this literature only inconsistently reports, and rarely reports jointly, is (i) an explicit statement of how a temporal or relational feature’s computation is bounded relative to the timestamp of the transaction it describes, and (ii) a same-architecture ablation isolating the graph component’s marginal

contribution over a tabular baseline of matched capacity. This paper’s own pre-rebuild history is direct evidence that this omission is not merely academic: without both a formal temporal-computation invariant and a first-class ablation, a reported gain from adding graph structure is not distinguishable, from the outside, from a gain produced by leakage or by an unmatched baseline. We return to this gap in Section II-H.

More recent surveys corroborate that this is a field-wide, not project-specific, gap. Cheng et al. [11] review the GNN-for-financial-fraud literature and find that leakage auditing and matched-baseline ablations remain the exception rather than the norm across the papers they survey, a conclusion echoed from the anti-money-laundering side by Deprez et al. [12], whose comprehensive review of continual graph learning for AML systems catalogs architectures and training regimes in detail but likewise finds few works that isolate structure’s marginal contribution under an explicit temporal-leakage guarantee. Lucas and Jurgovsky [13] and Chen et al. [14], surveying credit-card and broader financial fraud detection respectively, both note the same recurring pattern from the modeling side: papers report a single headline accuracy figure for a graph- or deep-learning-based detector far more often than they report the ablation needed to attribute that figure to structure specifically.

B. Money-Laundering Typologies and Cycle-Based Detection

A narrower thread within the graph-fraud literature targets the specific structural motifs – cycles, fan-in collection points, layered chains – this paper’s topology engine also detects. Weber et al. [15] released the Elliptic Bitcoin dataset and benchmarked graph convolutional networks against classical baselines for illicit-transaction classification, establishing graph topology as a first-class signal for cryptocurrency laundering detection, though on a dataset with a coarser, node-level (rather than this paper’s transaction-level) label. Li et al. [16] propose FlowScope, a dense-subgraph algorithm that explicitly targets multi-hop laundering flows through chains of accounts, and Tariq and Hassani [17] extend this line of work to billion-scale transaction graphs with a topology-agnostic detector aimed at temporal money-laundering flows specifically – both address the same class of structural motif this paper’s cycle and lapping detectors target, but neither reports a leakage-controlled, same-architecture comparison against a tabular baseline of matched capacity. Eddin et al. [18] instead target a downstream, operational problem this paper shares: reducing false positives in AML alert queues using graph-derived features, motivated by precisely the same auditor-facing constraint (a bounded daily review budget) that motivates this paper’s precision@ k reporting.

C. Class-Imbalance Learning

Fraud labels are rare by construction – across the four datasets used later in this paper, positive rates range from roughly one in a thousand transactions to just over one in a hundred – and imbalance-handling choices materially affect both accuracy and, importantly, its variance across training runs. Lin et al. [19] introduce focal loss for dense object

detection, reshaping the standard cross-entropy objective to down-weight easy, confidently-correct examples so that training gradient is concentrated on the hard cases; the same reshaping transfers naturally to rare-event tabular classification and is used later in this paper’s own model-selection protocol, compared head-to-head against simple class weighting under a rule that the two imbalance mechanisms are never combined within a single experiment. Zhi et al. [5], discussed above, instead address imbalance at the data level via conditional generative augmentation of the minority class within a graph-aware embedding space – a resampling family we deliberately avoid for structural (as opposed to purely tabular) features, since interpolating between two graph-derived feature vectors can produce a combination – for instance, half of a cycle – with no physical transaction-graph realization. Chawla et al.’s original SMOTE [20] is the ancestor of essentially every resampling-based imbalance technique used in this literature, including the generative variants surveyed above; Japkowicz and Stephen [21] and He and Garcia [22] independently document, across a broad range of classifiers and domains, that a resampling mechanism’s benefit is highly dataset- and feature-dependent rather than uniformly positive – the same caution this paper’s own methodology rule (one imbalance mechanism, never two, chosen and logged per experiment) is designed to operationalize rather than merely state. A 2026 alternative to resampling entirely is Wu et al.’s VAE-Inf [23], a statistically interpretable generative model for imbalanced classification that models the minority class’s distribution directly rather than interpolating between observed points; its motivation – that naive interpolation-based oversampling can manufacture statistically implausible synthetic examples – is structurally the same concern that leads this paper to exclude SMOTE from topology features specifically, applied there to a fully generative rather than nearest-neighbor-interpolation setting.

D. Temporal Graph Networks

Rossi et al. [24] introduce Temporal Graph Networks (TGNs): a memory-augmented architecture that maintains a compact, continuously-updated state vector per node, updated via a recurrent cell as each new edge (here, a transaction) is observed, and trained with a self-supervised link-prediction objective, via Adam [25], rather than direct supervision on any downstream label. This paper adopts a TGN, trained self-supervised and hence label-free, as the learned counterpart to the hand-crafted topology detectors in its offline research configuration: an account’s TGN embedding is read *before* the transaction that updates it is folded into memory, preserving the same temporal-ordering discipline used for the hand-crafted features, and the embedding is deliberately restricted from access to transaction amounts so that its contribution can be isolated from the model’s known amount-driven signal. Later sections of this paper are explicit, however, that this paper does not treat the TGN as free to deploy: floating-point non-determinism in CPU-trained recurrent updates compounds over an epoch’s worth of transactions to the point that TGN-derived embeddings retrained on the same, fixed random seed but on a different machine are not bit-reproducible against a

previously committed model, and the memory state itself is not persisted for streaming inference – both properties that keep the TGN a research-arm-only component in this paper’s accounting, never part of the production-served model. Related dynamic-graph architectures explore the same continuous-time setting from different angles: Wang et al. [26] learn inductive temporal representations via causal anonymous walks rather than a memory module, Dileo et al. [27] target temporal *heterogeneous* networks with a dedicated update-then-aggregate framework, and Yu et al. [28] propose a unified architecture and benchmarking library spanning several dynamic-graph-learning families – none of which this paper adopts directly, but all of which represent the same design space (a learned, continuously-updated node representation) that TGN occupies in this paper’s own ablation. A 2026 paper directly relevant to this design space is Lei et al.’s STC-MixHop [29], a multi-scale graph-learning framework with an explicit temporal-consistency constraint for fraud detection in non-stationary transaction networks; notably, Lei et al. report the same qualitative finding this paper’s own ablation arrives at independently – that “when node attributes are highly informative, tabular baselines remain difficult to outperform,” and that graph structure earns its keep specifically where relational patterns are operationally significant rather than universally. We return to this convergence when interpreting this paper’s own TGN-versus-topology comparison later in this paper.

E. Continuous Auditing and ERP-Centric Fraud Research

A parallel, older literature addresses fraud and control failures from the audit and information-systems side rather than the machine-learning side. Vasarhelyi et al. [30] study the acceptance and adoption of continuous auditing techniques by internal auditors, documenting that technique sophistication alone does not guarantee organizational uptake – relevant context for this paper’s emphasis on an audit console that surfaces not just a score but a reconstructed, human-checkable evidence path for every alert, since an unexplained score is exactly the kind of output that technique-adoption research finds auditors distrust. Jawad et al. [31] review machine-learning-driven optimization of ERP systems broadly, situating fraud scoring as one of several ML use cases layered on ERP data (alongside demand forecasting, process mining, and anomaly detection in master data) and noting that most such deployments remain post-hoc analytics layered outside the transactional system of record rather than integrated into it – precisely the gap this paper’s production console, built to score transactions at posting time rather than in a downstream batch job, is meant to close. Mamakou et al. [32] evaluate ERP systems post-implementation from internal auditors’ own perspective, underscoring that an ERP-adjacent tool’s practical value is judged by auditors against criteria – traceability, explainability of automated outputs, and compatibility with existing control frameworks – that a raw model accuracy number does not, by itself, satisfy. Together, this literature motivates why this paper treats the explanation and evidence-reconstruction layer as a first-class deliverable rather than an add-on to a scoring model.

Closer to the specific mechanics of ERP-native fraud detection, Tritscher et al. [33] release an open, simulated ERP dataset built specifically to enable reproducible occupational-fraud-detection research – addressing the same raw-data-availability gap that motivated this paper’s own `synth_erp` generator, though via a different game-based data-collection methodology. Schreyer et al. [34] detect accounting anomalies in a learned latent space via adversarial autoencoders, and Hemati et al. [35] extend this line of work with a continual-learning framework for unsupervised anomaly detection under the specific distribution shift that continuous, always-on auditing of real accounting data exhibits over time – both representative of an unsupervised, label-free alternative to this paper’s supervised approach, appropriate when confirmed fraud labels (as opposed to accounting irregularities more broadly) are not available. Foalem et al. [36] study logging requirements for continuous auditing of ML-based applications specifically, a systems-level concern this paper’s own structured-logging and correlation-id design (documented in the system section of this paper) independently arrives at from a production-engineering rather than an audit-research starting point.

F. Explainability for Tree and Graph Models

An accuracy figure, however well validated, does not on its own justify opening a fraud investigation; the audience for a fraud alert needs to know *why* a transaction scored highly. Lundberg and Lee [37] unify a family of additive feature-attribution methods under the Shapley-value-based SHAP framework, providing a game-theoretic, locally accurate decomposition of a model’s output into per-feature contributions. Lundberg et al. [38] extend this to TreeSHAP, an algorithm that computes exact (not sampled or approximated) Shapley values for tree-ensemble models in low-order polynomial time – a property this paper relies on directly, since it is a determining factor, alongside raw predictive accuracy, in this paper’s choice of a gradient-boosted tree ensemble as the final classifier head over an end-to-end neural alternative: exact, cheap attribution is available for trees in a way it is not for most deep architectures. On the graph side, Wang et al. [10] propose a reinforcement-learning-based causal explainer for GNN predictions, and Dao [39] more recently combines GNNExplainer with Shapley values for the same graph-native explanation problem, targeting a complementary explainability problem – explaining which subgraph structure drove a graph model’s own decision, rather than which tabular feature drove a downstream classifier’s decision. Almalki and Masud [40] instead stack SHAP, LIME, and permutation feature importance atop a gradient-boosted ensemble for fraud detection, an explainability-method combination closer in spirit to this paper’s own tree-native TreeSHAP choice than to the graph-native explainers above. This paper’s design deliberately sidesteps needing a GNN-native explainer of that kind for its production scoring path, feeding learned graph signal into TreeSHAP-explainable tree features rather than explaining a graph model’s decision directly; the graph itself is instead re-rendered for the auditor as the literal evidence path (the reconstructed cycle or fan-in structure a given alert

is attributed to), verified independently at display time against the same stored feature state the score was computed from.

Two 2026 papers make the regulatory stakes of this design choice explicit. Faccia [41] audits whether a high-performing fraud-detection model can simultaneously satisfy auditability, documentation, and regulatory-transparency requirements, comparing a weighted logistic-regression baseline against XGBoost with SHAP attribution under severe class imbalance and finding that exact, tree-native explanation is what actually closes the gap between predictive performance and audit-readiness – the same tradeoff this paper’s own choice of gradient-boosted trees over an end-to-end neural architecture is built around. Uddin and Aziz [42] evaluate explanation *reliability and consistency* across model families for fraud detection specifically because raw predictive accuracy is not, on its own, sufficient for regulatory sign-off, finding that XGBoost with TreeExplainer produces materially more stable explanations across repeated runs than LSTM-based alternatives – direct, independent corroboration of this paper’s own reliance on exact TreeSHAP rather than a sampled or model-agnostic approximation for every alert the production console surfaces.

G. Recent Advances in Graph-Based and Federated Fraud Detection (2025–2026)

Several works published in the twelve months preceding this paper extend the graph-fraud and financial-ML literature along directions adjacent to, but distinct from, this paper’s own leakage/ablation focus. On the throughput side, Ye et al.’s BlazingAML [43] compiles high-level laundering-pattern descriptions into optimized CPU/GPU code, reporting two- to three-hundred-fold speedups on the same IBM AML benchmark family used in this paper – a scalability contribution complementary to this paper’s own streaming, $O(1)$ -per-feature engine, targeting throughput rather than the leakage question this paper is built around. Premkumar et al. [44] survey AI-enhanced anti-money-laundering practice across financial institutions broadly, and Tian et al. [45] propose an adaptive GNN specifically for transaction fraud, both representative of the wider architecture space this paper deliberately narrows itself against (Section II-H). On the privacy and deployment side, Roth et al. [46] report an industry-scale evaluation of federated learning for cross-institution payment-fraud detection using NVIDIA FLARE, achieving federated performance that substantially closes the gap to a fully centralized model without sharing raw transaction data across institutions – directly relevant to this paper’s own single-tenant deployment scope, discussed explicitly as unevaluated future work later in this paper. Uddin et al. [47] extend financial fraud detection to a multilingual, Bangla-English setting using transformer models built on the self-attention mechanism of Vaswani et al. [48], a generalizability axis (language and locale, rather than dataset topology) this paper does not attempt to address but that is a natural complement to the dataset-diversity argument this paper’s own four-dataset roster makes. Finally, a broad base of applied credit-card and transaction-fraud studies – evaluating classical and deep architectures on imbalanced data [49], synthetic-data

generation via GANs for training augmentation [50]–[52], and causal, prototype-based approaches to realistic oversampling [53] – collectively motivate this paper’s own decision to treat synthetic-data augmentation as an empirical question to be tested rather than assumed beneficial, a question this paper answers directly, and largely negatively, later in this paper.

H. Positioning of This Work

Read across the sections above, the recurring shape of the gap this paper targets is consistent. Model novelty in the graph-based fraud literature is abundant: new architectures, new imbalance-handling mechanisms, new uncertainty-quantification and explainability layers all appear at a healthy pace, and the citations above are one small, recent sample of that pace. What is comparatively scarce, and what this paper is built around supplying, is methodological accounting for two specific threats to validity that are easy to state and easy to violate silently: whether a temporal or relational feature could, even in principle, have depended on information from after the event it describes, and whether an architecture’s graph component is being compared against a tabular baseline of genuinely matched capacity rather than a weaker straw-man. Neither threat is exotic – this paper’s own preliminary iteration fell to both simultaneously, with a future-looking feature ranked among its most important, before an internal audit caught it – and neither is fully addressed by adding a leakage *test* after model development, since a test can only catch a violation the engineer thought to check for. The response this paper takes is architectural rather than remedial: a read-then-insert invariant enforced in the feature engine itself (Section I), guarded by an automated regression test rather than a one-time audit, and a within-dataset, same-architecture ablation run identically across four structurally different datasets rather than a single headline comparison. This paper’s technical components – hand-crafted topology detectors, a TGN [24], gradient-boosted trees, focal-loss and class-weight imbalance handling [19], and TreeSHAP attribution [38] – are consequently adaptations of established techniques rather than new architectures in their own right. The contribution is the discipline wrapped around them, and the willingness, evidenced throughout this paper, to report what that discipline finds even when it is a negative result on the very dataset the project’s original hypothesis was built to explain.

III. PROPOSED MODEL AND METHODOLOGY

A. Problem Formulation and Leakage Taxonomy

1) *Task Definition*: We formalize fraud detection on ERP and payment data as per-transaction binary classification over a temporal, directed, attributed multigraph. Let $\mathcal{V}$ denote the set of entities (accounts, vendors, G/L codes, or customers, depending on the dataset) and let $\mathcal{D} = \{e_1, e_2, \dots, e_N\}$ denote the transaction set, where each transaction is a tuple

$$e_i = (u_i, v_i, t_i, a_i, \mathbf{c}_i, y_i),$$

with origin $u_i \in \mathcal{V}$, destination $v_i \in \mathcal{V}$, timestamp $t_i \in \mathbb{R}$ (or an ordinal step/day index – Section III-B details each dataset’s time axis), amount $a_i \in \mathbb{R}_{>0}$, an auxiliary attribute vector

$\mathbf{c}_i$ (transaction/document type, channel), and a binary label $y_i \in \{0, 1\}$ recording confirmed fraud. Transactions induce a directed multigraph $G = (\mathcal{V}, \mathcal{D})$ that grows monotonically in time; we write

$$G_{\leq T} = (\mathcal{V}, \{e_j \in \mathcal{D} : t_j \leq T\})$$

for the graph state immediately after every transaction up to and including time T has posted. The task is to fit a scoring function $f_\theta : \mathbb{R}^d \rightarrow [0, 1]$, applied to a feature vector $x_i = \phi(e_i, G)$ extracted for each transaction, that approximates $\Pr(y_i = 1 \mid x_i)$. Three feature families compose x_i in this project: ordinary transaction attributes (a_i , $\mathbf{c}_i$, calendar features), structural/topology statistics computed from the graph (degree, fan-in/out, lapping recurrence, cycle membership), and, where used, learned temporal-graph embeddings (TGN) [24]. The graph neural network literature has repeatedly reported structure-aware models outperforming tabular baselines on financial fraud tasks [1], [2], almost always without an explicit audit of the temporal-leakage failure modes formalized below; the dataset roster in Section III-B is chosen so that a fair, leakage-controlled version of that comparison is possible across a genuine range of structural richness. With the class prior $\Pr(y = 1)$ ranging from roughly 0.12% to 1.21% across the real datasets studied here (Table II), we follow standard practice for severely imbalanced classification [54] and evaluate with threshold-independent ranking metrics (PR-AUC / average precision, precision@ k) rather than accuracy or a fixed decision threshold; the full metric discussion and the modeling protocol built on it are given later in this paper.

2) *Leakage-Free Feature Computation*: The single methodological requirement that dominates every other design decision in this project is that ϕ must not use information unavailable at the instant e_i posts. We state this precisely.

Definition (leakage-free feature map). A feature map ϕ is *leakage-free* with respect to $\mathcal{D}$ if, for every transaction $e_i \in \mathcal{D}$,

$$\phi(e_i, G) = \phi(e_i, G_{\leq t_i})$$

for any $G \supseteq G_{\leq t_i}$ – that is, ϕ ’s output for e_i is invariant to which transactions with $t_j > t_i$ are or are not present in the graph.

Operationally we enforce this with a single streaming pass over $\mathcal{D}$ in non-decreasing timestamp order: for each e_i , (1) *read* $x_i = \phi(e_i, G_{< t_i})$ against the graph state already committed, then (2) *insert* e_i ’s edge and any derived state (running counts, expanding means, open cycle candidates) before moving to e_{i+1} . This read-then-insert ordering makes look-ahead structurally impossible rather than merely policy-forbidden – no code path exists through which a later transaction’s data can reach an earlier transaction’s feature vector. Two corollaries follow directly and are enforced throughout the engine: (a) no statistic used in a feature computation – mean, standard deviation, per-account baseline, Benford digit distribution – may be computed once over the whole dataset; every such statistic must be expanding or rolling over the past only, and (b) when a structure (a cycle, a lapping sequence) is only detectable once its final edge posts, its signal attaches to

that *completing* transaction, never retroactively to the earlier transactions that merely participated in it. Train/validation/test partitioning follows the same discipline: splits are chronological – train on the earliest period, validate on the next, test on the latest – never a randomly shuffled k -fold, a protocol documented elsewhere to silently leak the future into the past on time-evolving, dynamic fraud data [55].

3) *Leakage Taxonomy: Two Failure Modes in the Predecessor System*: An earlier iteration of this codebase reported headline metrics as strong as ROC-AUC ≈ 0.997 on its flagship dataset and concluded that graph topology was fraud detection’s “ultimate differentiator.” A pre-registration audit of that system’s own artifacts found the opposite: its topology detectors scored essentially 0.0 feature importance, and every reported number was contaminated by two independent, stacked leakage mechanisms. We name them Leak Class A and Leak Class B and treat their elimination as this project’s primary internal-validity requirement.

Leak Class A – whole-graph feature computation. The predecessor’s topology detectors ran once over the *entire* graph, with no $t_j \leq t_i$ restriction anywhere, and each detected structure’s signal was written back to every member transaction, including its earliest ones – a cycle only observable once closed was nonetheless flagged on the edge that opened it, directly violating the definition above. Two concrete symptoms make the violation unambiguous: (i) a feature literally named `time_to_next` – the number of seconds until each account’s *next* transaction – ranked among the top five features by importance; no detector can know the gap to a future transaction without having already observed it. (ii) Every z-score / anomaly feature was standardized against a whole-dataset mean and variance computed once, so a transaction dated on day one was scored against a “normal” baseline that included transactions from every later day, test period included.

Leak Class B – modeling-protocol leakage. Four compounding, confirmed malpractices were present in all five of the predecessor’s training scripts. (i) SMOTE over-sampling and class-weighting were applied *simultaneously*, double-correcting for imbalance; because SMOTE’s k -nearest-neighbor interpolation was applied indiscriminately to topology columns, it manufactured physically impossible points (a synthetic transaction with, e.g., 0.4 of a closed cycle) that eased the learned boundary without corresponding to any real fraud pattern. (ii) Every reported metric used the library’s default 0.5 decision threshold (`.predict()`), never a threshold selected on held-out data – meaningless when the positive class is 0.1–1.2% of rows, since 0.5 is not an operating point anyone chose. (iii) Cross-validation used `StratifiedKFold` with shuffling and the timestamp column dropped before fitting, so folds routinely trained on transactions dated after the ones being scored, layering random-future leakage directly on top of Leak A. (iv) Models were selected by F1 at that same fixed 0.5 threshold rather than a threshold-independent, imbalance-appropriate ranking metric, which rewards whichever leak-inflated configuration happens to look best at an arbitrary cutoff rather than the one that generalizes.

Why each invalidates the reported numbers. Leak A

This paper’s engine: read-then-insert



Predecessor system: whole-graph attribution (leaky)



Fig. 2: Read-then-insert feature computation (top) versus the predecessor system’s whole-graph attribution (bottom), which credited a structure’s signal back to transactions that occurred before the structure existed.

inflates apparent predictive power because the model is not learning what is knowable at the moment an investigator would act – it is learning the correlation between a label and events that, by construction, occur after the label’s determining moment; the resulting metric is not an estimate of deployable detection ability at all. Leak B compounds this at the evaluation layer: synthetic minority points generated from a distribution informed by the whole, undifferentiated dataset bias the decision boundary toward the very examples later used to measure it; shuffled folds directly violate the temporal ordering a fraud investigation depends on; and selecting the “best” configuration by a metric computed under both prior leaks means every design choice in the pipeline was made with implicit knowledge of the answer it would later be graded on. Under datasets this imbalanced – as few as 19–25 confirmed fraud transactions in a single evaluation split for SAP Würzburg (Section III-B) – even a handful of leaked examples can move a headline metric by tens of points, which is precisely the failure mode the leakage-free construction above, and the ablation protocol built on it, are designed to make structurally impossible rather than merely audited for after the fact.

B. Datasets

1) *Overview and the Topology-Richness Spectrum*: Because the hypothesis under test is whether *graph structure* carries predictive signal beyond a strong tabular baseline, the dataset roster is presented here as points along a topology-richness spectrum rather than as an arbitrary convenience sample: from IBM AML, whose sender and receiver identifiers occupy a near-unified account namespace supporting true multi-hop cycles, down to PaySim, whose transaction graph is a near-forest of one-off pairs. Table II summarizes the four datasets computationally profiled end-to-end in this study – IBM AML, BankSim, SAP Würzburg, and PaySim – drawn from a single calibration pass over each dataset’s fully processed transaction table (`calibration_reference.json`) [56]–[58]. *Kind* records each dataset’s native graph topology: *Transfer* (a directed account→account graph, as in a payments network), *Bipartite* (two structurally disjoint node classes with no possibility of reciprocation or cycles), or *ERP ledger*

TABLE II: Graph-topology and scale summary of the four real datasets computationally profiled in this study (source: `calibration_reference.json`). Amounts are in each dataset’s native units and are not comparable across datasets.

Dataset	Kind	Fraud %	N_{trans}	N_{acc}	N_{acc}	Amt. Med.	Benford MAD	NS Overlap	Recip.
IBM AML	Transfer	0.130	1,719	1,323,234	9,999	156.71	0.0758	0.9927	0.0446
BankSim	Bipartite	1.211	7,200	594,643	4,162	26.90	0.0319	0.0000	0.0000
SAP Würzburg	ERP ledger	0.124	248	200,629	59,883	1,975.80	0.0385	0.0000	0.0000
PaySim	Transfer	0.129	8,213	6,362,620	9,073,900	74,871.94	0.0142	0.0002	0.0000

(the document-to-ledger-account bipartite structure specific to double-entry ERP postings). *NS Overlap* is the fraction of an account namespace’s senders that also appear as receivers – the diagnostic this project uses to certify whether same-entity cycles are even structurally detectable; the predecessor system failed exactly this check by splitting senders and receivers into disjoint node types, making a loop $A \rightarrow B \rightarrow A$ structurally invisible regardless of how good its topology engine was. *Reciprocity* is the fraction of unique (sender, receiver) pairs for which the reverse pair also transacted. Of the four datasets in Table II, only PaySim was profiled without being modeled (Section III-B6).

2) *IBM AML*: IBM AML is generated by the AML-Sim multi-agent simulator and released as a synthetic-but-structurally-realistic anti-money-laundering benchmark [56]. Its 1,323,234 transactions connect 9,999 accounts with a sender/receiver namespace overlap of 99.27% – the same account IDs act as both senders and receivers almost everywhere, the pre-condition this project treats as necessary for account-to-account cycle detection to be structurally possible at all. Every transaction carries the type `TRANSFER`; time is an integer day index over a 199-day span. At a fraud rate of 0.130% (1,719 of 1,323,234), IBM AML is this project’s richest-structure testbed and the dataset on which the topology hypothesis is given its best chance to succeed. It also exhibits the sharpest amount anomaly of any dataset studied: fraud amounts have median 10.60 and mean 9.76, against a full-population median of 156.71 and a mean of 115,988.18 (99th percentile 1,854,724.69) – fraud in IBM AML clusters in a narrow, near-zero-value band while the legitimate population is heavily right-skewed toward large transfers. We read this as an artifact of the AMLSim generator’s labeling convention rather than a general property of laundering behavior, and it materially shapes the ablation results discussed later in this paper: an amount-based baseline that expects fraud to look large is directionally wrong on this dataset, which is exactly what motivates evaluating an amount-blind probe to isolate topology’s contribution independent of this artifact.

3) *BankSim*: BankSim is an agent-based simulation of a bank payment network built specifically for fraud-detection research [57]. Its 594,643 transactions connect customer accounts (prefix `C`) to merchant accounts (prefix `M`) – two structurally disjoint node classes, so namespace overlap and pair reciprocity are both exactly 0.0 and no customer-to-customer or merchant-to-merchant cycle can exist by construction. BankSim therefore serves this study as the sparse control: any predictive lift topology features provide here must come from bipartite-graph statistics (degree, fan-in/out, recurring near-equal “lapping” amounts) rather than from cycles, since

cycles are structurally unavailable. Its fraud rate, 1.211% (7,200 of 594,643), is the highest of the four real datasets, and its amounts are the smallest in scale (median 26.90).

4) *SAP Würzburg*: SAP Würzburg is a real, publicly released SAP ERP transaction ledger and is the only dataset in this study drawn from an actual production accounting system rather than a simulator – the setting continuous-auditing research argues automated, always-on control monitoring should target [30], and one that broader ERP-analytics surveys still treat as an emerging rather than mature application of machine learning [31], [32]. Its graph is a document-to-account bipartite structure native to double-entry bookkeeping: each posting document fans out to a small number of line items (out-degree median 2.0, 99th percentile 10.0) that land on accounts, a minority of which absorb a disproportionate share of postings (in-degree 99th percentile 43,706.7, maximum 46,554, against 59,883 distinct account identifiers in total) – a small core of heavily used ledger accounts against a long tail of low-traffic counterparty accounts. With only 248 confirmed frauds among 200,629 postings (0.124%), SAP Würzburg is this project’s thinnest fraud signal by count, and, as the ablation study reported later in this paper shows in full, the one real dataset on which the full-model topology lift is measured *negative*. Two different SAP Würzburg train/validation/test partitions appear elsewhere in this paper and should not be confused: the chronological, partition-aware split used in the within-dataset ablation study yields 120,411/40,137/40,139 rows (`ablation.json`), while the multi-seed split used for final champion-model selection yields 120,341/40,114/40,116 rows on the same source file under a different, non-partition-aware split procedure (`final_champions.json`); both carry the same fraud counts by split (204/19/25). We flag the discrepancy here so it does not appear silently across two disconnected tables.

5) *IBM AML: Kaggle Reproduction Variant*: A second, independently downloaded instance of the IBM AML benchmark – sourced directly from its Kaggle mirror rather than from the AMLSim-format repository export behind the headline IBM AML numbers above – was processed during this build as a reproduction check on the deployed *online* scoring bundle, never as a source of headline results. The two sources use different raw schemas (`TX_ID/SENDER_ACCOUNT_ID`-style AMLSim export versus the Kaggle `HI-Small_Trans.csv` layout) and are, in general, different simulator draws; we report results under the separate label `ibm_aml_kaggle` throughout, and none of its numbers are merged into or averaged with `ibm_aml`’s. For contrast – though the two are not directly comparable – the offline champion model trained on the primary `ibm_aml` source, which includes TGN embeddings and, since its TGN memory state is never persisted to disk, is not servable online at all (a limitation discussed in full elsewhere in this paper), reaches a saturated test PR-AUC of 1.00; that ceiling is itself uninformative given the baseline’s own 0.990 saturation (also discussed later in this paper). `ibm_aml_kaggle`’s *online* (ordinary+topology, non-TGN) bundle – the only kind of bundle this project ever serves in production (`online_bundles.json`) – reaches a materi-

ally lower test PR-AUC of 0.512 (± 0.067 over three seeds, 26 features), consistent with both the missing TGN component and the fact that the two `ibm_aml` variants are different simulator draws, not repeated measurements of the same result.

6) *PaySim: Profiled, Not Modeled*: PaySim simulates a mobile-money transfer network at substantially larger scale than any other dataset in this study – 6,362,620 transactions over 9,073,900 accounts [58], where almost every transaction is the only time its particular (sender, receiver) pair ever transacts (namespace overlap 0.0002, zero pair reciprocity, zero repeat-pair transaction fraction). Its fraud rate, 0.129% (8,213 of 6,362,620), sits in the same narrow band as IBM AML and SAP Würzburg, but its graph is a near-forest at the opposite end of the topology-richness spectrum from IBM AML: with essentially no repeated edges between the same two accounts, most multi-hop structural detectors (cycles, recurring lapping, degree accumulation) have little to attach to before a transaction resolves. Consistent with this project’s tiered dataset plan, PaySim was carried through the ETL and topology-profiling stages reported in Table II but was never carried into model training or the ablation study reported later in this paper; no PR-AUC, precision@ k , or lift number exists for it anywhere in this work, and its row in Table II should be read purely as a structural reference point completing the spectrum, not as a modeling result.

7) *The Synthetic Economy*: `synth_erp`: `synth_erp` is this project’s own agent-based synthetic ERP economy, spanning 39,256 accounts, generated deliberately *after* the real-data ablation protocol was designed, specifically to avoid the circularity risk of injecting fraud in the exact shape a detector is built to find – a trap the predecessor system fell into by labeling injected fraud literally "Circular Payment" or "Lapping". Fraud is instead generated from simulated criminal intent and evasion behavior, and an independent adversarial leakage probe run against the resulting table reports no exploitable single- or two-feature giveaway rule (`leak_flag`: false, `synth_erp_realism.json`); the worst rule found, `tx_type=transfer & dest_created_midear`, reaches only 9.29% precision at 43.75% recall of fraud, a $19\times$ lift over the 0.488% base rate that is unremarkable next to a genuinely leaking rule. At 1,148,503 transactions and a fraud rate of 0.488% – deliberately elevated above any real dataset’s 0.124–1.211% range to keep minority-class training stable – `synth_erp` was built with a near-unified account namespace (0.941 overlap) and the highest pair reciprocity of any dataset in this study (0.136), by design, to give topology detectors rich structure to find. Its amount distribution is, if anything, *too* clean relative to real data: a Benford MAD of 0.0047 is roughly three to sixteen times lower than any real dataset’s (0.0142–0.0758, Table II) – more compliant with Benford’s law than any of the messy real ledgers it is meant to approximate, a known limitation of amount generation in synthetic financial data that we surface rather than smooth over. Per the project’s own circularity safeguard, `synth_erp` is used only for training-time augmentation and stress-testing experiments, reported later in this paper; it is

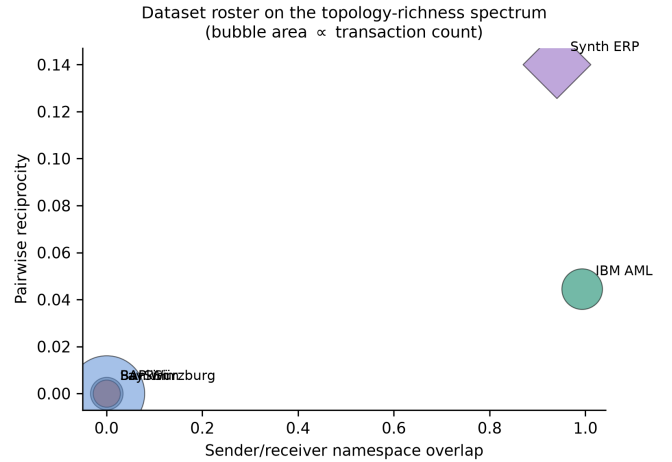


Fig. 3: The five datasets positioned on the two structural axes this study uses to justify the roster: sender/receiver namespace overlap (x) and pair reciprocity (y), bubble area proportional to transaction count. IBM AML and `synth_erp` anchor the structure-rich end; BankSim, SAP Würzburg, and PaySim anchor the structurally sparse end.

never a source of this paper’s headline ablation claim, which is evaluated exclusively on the three real, independently sourced datasets in Table II that were carried into modeling, plus SAP Würzburg’s ERP-native structure specifically.

C. Leakage-Free Temporal Topology Engine

A transaction does not reveal fraud in isolation; the same transfer can be ordinary in one account context and the closing leg of a laundering cycle in another. Graph-based fraud detection formalizes this by treating the ledger as a network of who-paid-whom and searching for structural motifs — cycles, fan-in collection points, rapid pass-through chains — that a row-wise view cannot see [1], [2], [59]. The difficulty is computing such motifs *as they would have been visible at the time*, without letting the completed, labeled dataset’s hindsight leak into a feature that is supposed to represent an auditor’s knowledge at transaction time. This section describes the streaming component of the Self-Auditing Ledger that does this — the Temporal Topology Engine — and states, precisely and with the project’s own evidence, which of its thirteen output features actually earn their keep.

The engine operates on the canonical edge list produced by ETL: one row per transaction, with a source account, a destination account, a timestamp, an amount, and (for training data) a fraud label. It is deliberately *not* a graph neural network and does not touch Neo4j; it is a single linear pass over the sorted edge stream, maintained in pure Python/NumPy, that emits one feature vector per transaction. Two data structures carry all state across that pass: a trailing-window directed multigraph, `WindowGraph`, and a per-account rolling amount memory, `LapMemory`. Both are described below, followed by the thirteen features they jointly produce, the structure/amount split used to keep the later ablation honest, and the feature-

importance result — including a null result the project does not soften.

1) *The Leakage Invariant: Read, Then Insert:*

CLAUDE.md’s rule 4 states the requirement in one sentence: a transaction’s features may depend only on transactions with `timestamp <= its own`. The engine satisfies this not as a coding convention that a future edit could quietly violate, but as a structural property of a single loop body. Edges are pre-sorted by `(timestamp, transaction_id)` — the ETL writes them in this order with a stable sort, and the feature-assembly stage re-sorts defensively if that invariant is ever violated upstream. The engine then iterates for `i` in `range(n)` and, for transaction `i`, executes exactly two kinds of statement in exactly one order:

- 1) **Read.** Query `WindowGraph` for `i`’s fan-in, fan-out, destination in-degree, source out-degree, and (if `srci ≠ dsti`) run the bounded cycle search; query `LapMemory` for the recurrence of `i`’s amount against each endpoint’s rolling history.
- 2) **Insert.** Only after every read above returns does the engine call `WindowGraph.add(t, u, v, amount)` and `LapMemory.record(...)` for transaction `i` itself.

Because `add` for edge `j` is called only during iteration `j`, and iterations execute strictly in order `0, 1, ..., n-1`, the graph state visible to iteration `i`’s reads is, by induction, exactly $\{j : j < i\}$ minus whatever `evict` has aged out — never a superset, and in particular never anything from an iteration that has not yet run. No branch of the loop calls `add` or `record` ahead of the paired reads for the same `i`, and `evict` only removes edges whose timestamp falls below `now - window`; it has no code path that admits an edge, only ones that retire old ones. This is what we mean by *leakage-free by construction*: the claim does not rest on trusting that every caller remembers to query before mutating, or on empirically observing no leak on the datasets at hand — it follows from the fixed order of two statements inside one loop, for any input that satisfies the pre-sort precondition. `LapMemory` enforces the identical discipline: `recurrence(acct, amount)` is always read before the same call site’s `record(acct, amount, counterparty)`. For SAP Würzburg, whose synthetic per-run time axis (encoded as `run_index × 106 + seconds-since-midnight`) is not a real cross-run chronology, both structures are additionally discarded and rebuilt at every `run_id` boundary, so a document in run 214 can never borrow structural context assembled from a different, unrelated ledger run.

This constructive guarantee is checked, not merely asserted: an automated leakage test (`topology.leakage_test`) re-derives the same feature table twice on real, built data — once truncated at the halfway point, and once with every transaction *after* the halfway point corrupted (shuffled endpoints, amounts overwritten with an out-of-range sentinel) while timestamps and row order are preserved — and asserts the first half’s features are byte-identical in both cases. This is deliberately redundant with the constructive argument above: the constructive argument is the proof that leakage is impossible for any input; the test is a regression guard that would fail loudly if a

future refactor moved a read after an insert. It has run green throughout the build.

2) *The Trailing-Window Graph:* `WindowGraph` holds a directed multigraph restricted to a trailing time window $[\text{now} - w, \text{now}]$, where `w` is a per-dataset constant (7 days for IBM AML; 7 steps for BankSim, whose control role requires `in_cycle` to be identically zero everywhere since its graph is customer→merchant bipartite; 24 steps = 1 day for PaySim; and unbounded *within* a run for SAP Würzburg, whose runs are short enough that no window truncation is needed). Internally it keeps adjacency as per-node dictionaries of neighbor multiplicities, so fan-in/fan-out (count of *distinct* counterparties) and weighted in-/out-degree (count *with* multiplicity) are both $O(1)$ lookups against the current window state, and a deque of `(timestamp, u, v)` triples in arrival order lets `evict(now)` pop expired edges from the front in amortized $O(1)$ per evicted edge. The one non-trivial query is the cycle search: does inserting $u \rightarrow v$ close a directed loop $v \rightsquigarrow u$ already present in the window? This is answered by a breadth-first search from `v` along existing out-edges, capped at `max_cycle_len-1` hops (default 6) and a visited-node `search_budget` (default 4,000); either bound being hit returns “no cycle” rather than risk a false positive or an unbounded cost on a dense hub account, so the detector is a conservative under-approximation of the true cycle set, never an over-approximation. Each directed pair also remembers the `(amount, timestamp)` of its most recent in-window edge, which the cycle search folds together to report amount conservation and timing tightness around the closed loop without a second graph pass.

3) *Per-Account Lapping Memory:* Lapping — using a new, similarly sized inflow to cover an earlier shortfall, “robbing Peter to pay Paul” — is not naturally a graph-window feature: its tell-tale recurrence can be separated by weeks or months, far longer than any window size that keeps the cycle search cheap. `LapMemory` instead keeps a fixed-length, *count*-based rolling history of the last $k = 25$ (amount, counterparty) pairs an account was involved in, in a given role (incoming or outgoing) — two independent instances are kept per account, one for each role. Because eviction is by count, not by elapsed time, this memory spans exactly as much calendar time as the account’s own activity took to accumulate 25 transactions, so a dormant account that reappears after a long gap is still compared against its own last known pattern rather than an empty, window-truncated one. `recurrence(acct, amount)` scans that account’s stored history for entries within a relative tolerance `tol = 1%` of the query amount and returns both the count of matches and the number of *distinct* counterparties among them. The second number is what makes this a lapping detector rather than a recurring-bill detector: one counterparty repeating the same payment every month is ordinary billing, but several *different* counterparties each supplying a near-identical amount to the same account is the diversification pattern lapping schemes exhibit. Reporting raw recurrence count alone would fall into the same triviality trap as raw cycle membership (§IV-A) — it fires constantly on legitimate recurring payments — so counterparty diversity is carried as a co-equal, separate feature rather than folded into

a single scalar.

4) *The Thirteen Topology Features*: Every transaction receives the same thirteen columns, grouped below by the structure they measure; all are computed from the past-only graph/memory state exactly as described above.

Cycle detectors (5). `in_cycle` is 1 if the transaction closes some directed loop within the window (a same-account self-payment counts as a degenerate length-1 loop and is flagged directly, without a search); `cycle_length` is the edge count of the *shortest* such loop found, 0 if none; `in_cycle_ge3` restricts this to loops of length ≥ 3 , excluding both self-loops and simple reciprocal pairs ($A \rightarrow B \rightarrow A$), which are common and largely benign; `cycle_amount_ratio` is $\min/\max$ of the amounts around the closed loop (including the closing edge itself) — near 1 means the money round-tripped with its value intact, the discriminating signature of a wash trade rather than, say, a sale followed by a small partial refund; `cycle_time_span` is the closing transaction’s timestamp minus the earliest edge on the loop — small values mean the loop closed quickly, the “tightness” that separates a fast laundering round-trip from an incidental, slow-forming one. Both are NaN when no cycle closes, left as missing for the tree model rather than imputed to zero.

Degree/fan detectors (4). `fan_in` and `fan_out` count, respectively, the number of *distinct* accounts that have sent to the destination account and the number of *distinct* accounts the source account has paid, within the window; `dest_in_degree` and `src_out_degree` count the corresponding edges *with* multiplicity. The distinct/multiplicity pairing lets the model separate “many different counterparties” (structuring, collection points) from “one counterparty transacting repeatedly” (high degree, low fan) — the same diversity-versus-repetition distinction the lapping features apply to amounts.

Lapping detectors (4). `lap_in_recur` and `lap_in_payers` are `LapMemory.recurrence` evaluated against the destination account’s *incoming*-amount history; `lap_out_recur` and `lap_out_payees` the same against the source account’s *outgoing*-amount history (§III-C3).

5) *Structure vs. Amount: Why the Split Matters*: Of the thirteen features, five are computed *from transaction amounts* — `cycle_amount_ratio` and all four lapping features — while the remaining eight (`in_cycle`, `cycle_length`, `in_cycle_ge3`, `cycle_time_span`, `fan_in`, `fan_out`, `dest_in_degree`, `src_out_degree`) are pure graph shape and never look at an amount value. `cycle_time_span` is time-derived, not amount-derived, and is counted with the structure group despite superficially “sounding” adjacent to the amount features. The engine exposes this split (`topology_structure_cols` vs. `topology_amount_cols` in the feature manifest) because the ablation study’s central probe — does topology help *independent of the amount signal*? — requires dropping every ordinary amount-derived tabular feature (`log_amount`, the expanding amount z-score, etc.) *and* every amount-derived topology feature together. Running an “amount-blind” arm that strips the ordinary amount columns but leaves `lap_*` and

`cycle_amount_ratio` in place would silently re-admit the amount signal through the back door and overstate what pure graph shape contributes. The structure-only arms in Table V’s source data (`no_amount_plus_structure` in `paper/data/ablation.json`) are the clean version of this test; their interpretation is developed fully in the ablation study.

D. Modeling and Evaluation Protocol

Every number reported in this paper and the next is produced by a single, shared training-and-evaluation harness applied identically to all four modeled datasets (BankSim, IBM AML, SAP Würzburg, and the in-house `synth_erp` corpus). This section specifies that harness precisely enough to be reproduced: the learner and its fixed hyperparameter defaults, the two mutually exclusive imbalance mechanisms, the chronological splitting rule (including a partition-aware variant required by one dataset’s own construction), the threshold-freezing procedure, the metric suite, and the attribution method. The harness is deliberately narrow — one function, one set of rules, applied the same way whether the feature set under test is a lean eight-column ordinary baseline or a ninety-nine-column champion bundle. Section IV-B then uses this exact protocol, unmodified, to answer the paper’s central question.

1) *Model family*: *gradient-boosted trees*, *histogram method*: Every arm in this paper is an XGBoost classifier [60] with `tree_method="hist"`, trained for up to 500 boosting rounds with early stopping after 40 rounds without validation-PR-AUC improvement. A single locked default configuration anchors every comparison: `max_depth=6`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, `min_child_weight=5`, `reg_lambda=1.0`, `reg_alpha=0.0`. Where a light random hyperparameter search (20 sampled configurations, scored on validation PR-AUC only) is run for a dataset’s final champion, this default configuration is always included as candidate zero, so a tuned model can never underperform the untuned baseline on the selection metric it will be judged by. The search is automatically skipped when a split’s validation-set fraud count is judged too small to support it without overfitting hyperparameters to a handful of positives — the guard that fires for SAP Würzburg (19 validation frauds), discussed further in §IV-B2.

Two other model families were deliberately not adopted as the classifier head, for reasons specific to this paper’s ablation design rather than general performance claims. LightGBM [61] and Random Forest [62] are both viable gradient-boosted or bagged-tree alternatives to XGBoost and would very likely be TreeSHAP-compatible in the same way; XGBoost was retained throughout purely for harness consistency with this project’s earlier committed artifacts, not because of a documented advantage over these alternatives, and re-running the ablation on either is named as future work rather than assumed unnecessary. Graph-native architectures — Kipf and Welling’s GCN [63], Veličković et al.’s GAT [64],

and Hamilton et al.’s GraphSAGE [65] (see also Hamilton’s own textbook treatment of the broader graph-representation-learning field [66]) — were considered and not adopted for a different, more consequential reason: none offers, out of the box, an attribution mechanism as cheap and exact as TreeSHAP’s polynomial-time traversal of a fitted tree ensemble, and this paper’s explainability requirement (every alert gets an exact, not sampled, per-feature attribution) is treated as a hard constraint on the model family, not a feature added after the fact. Lei et al.’s 2026 STC-MixHop [29] is a recent graph-native architecture aimed at exactly this paper’s problem setting (non-stationary transaction fraud), and its own reported finding — that a graph-native model struggles to outperform a tabular baseline precisely where node attributes already carry strong signal — is consistent with, and independently motivates, this paper’s decision to route learned graph signal (via TGN) into TreeSHAP-explainable tabular features rather than deploy a graph-native model end to end.

2) *One imbalance mechanism, never two*: Fraud rates in the four datasets range from roughly 0.1% to 1.2% of rows. The harness offers exactly two ways to counteract this, selected per run and recorded in every saved artifact, and never combined: (i) *class weighting*, XGBoost’s native `scale_pos_weight` set to the train-split negative-to-positive ratio, and (ii) an *alpha-free focal-loss* objective [19], implemented as a custom XGBoost objective with a closed-form gradient and a central-finite-difference Hessian, trained with `scale_pos_weight=1.0`. The focal variant is deliberately *unweighted* (symmetric, no per-class α term): the common alpha-balanced formulation of focal loss secretly reintroduces a class-prior weight, which would silently stack a second imbalance mechanism on top of the first — precisely the failure mode this protocol is built to exclude. Because a focal objective returns a raw margin rather than a probability, every downstream consumer (threshold tuning, PR-AUC, TreeSHAP) is written to accept either representation, applying a sigmoid only where a probability is actually required. Synthetic minority oversampling (SMOTE) is excluded from the harness entirely, on the grounds that an interpolated point between two topology vectors — half a cycle, a fractional fan-in count — has no physical meaning in a transaction graph.

Which mechanism wins is decided the same way as any other hyperparameter: on validation-split PR-AUC, never on the test split. Per `final_champions.json`, focal loss wins the selection on three of the four datasets — BankSim (validation PR-AUC mean 0.9515 vs. 0.94891 for class weighting), SAP Würzburg (0.35619 vs. 0.28875, though this particular selection is separately flagged `reliable: false` owing to the small-sample guard above), and `synth_erp` (0.85066 vs. 0.84866). IBM AML is the exception, and an instructive one: both mechanisms reach a validation PR-AUC of exactly 1.0 with zero variance across three seeds, so the “winner” (class weighting, by tie-break) is not really a comparison at all — an early symptom of the baseline-saturation story developed in §IV-B.

3) *Chronological splitting, including a partition-aware variant*: Every split is a 60/20/20 train/validation/test partition by transaction timestamp, and rows are never shuffled: the

training set is defined as the earliest 60% of transactions by time, validation the next 20%, and test the final, strictly future 20%. For most datasets this is a single global quantile cut on the timestamp axis. SAP Würzburg is the exception: its raw export orders postings by `run_id` (audit run), and each run is internally fraud-first-then-clean, so a global time cut would strand an entire zero-fraud run inside the validation split. The harness therefore applies a *partition-aware* variant for this dataset — the same 60/20/20 chronological rule, but computed independently within each `run_id` and concatenated, so every run contributes its own early rows to train and its own late rows to test. This is still a strictly chronological, never-shuffled split; only the unit over which the time cut is computed changes.

Two independent partition-aware splits of SAP Würzburg appear in this project’s own committed artifacts, and they differ by a small margin that we flag rather than silently reconcile. The ablation run (`ablation.json`) reports train/val/test sizes of 120,411/40,137/40,139; the later final-champion run (`final_champions.json`) reports 120,341/40,114/40,116 — a difference of 116 rows, or about 0.06% of the dataset. Both runs use the identical partition-aware algorithm described above, so the discrepancy is not a change in splitting logic; it most plausibly reflects a small difference in row count between two separate ETL/feature-build passes of the same source file. We did not further trace its exact origin. Every SAP Würzburg figure in this paper is therefore attributed to the specific source file it was read from, and the two split-size counts should not be added, averaged, or treated as interchangeable.

4) *Threshold tuning: frozen before the test set is touched*: XGBoost outputs a continuous score; converting it into a binary alert requires a cutoff, and an arbitrary 0.5 is never used. Instead, the harness scans the validation-set precision–recall curve and selects the threshold that maximizes F1 on validation data alone. That threshold is then frozen and applied, unmodified, to the held-out test split — the test set never participates in threshold selection. In practice the chosen threshold varies considerably by feature set even within one dataset (for BankSim, 0.9673 for the ordinary baseline vs. 0.9549 for the topology-augmented model; `ablation.json`), underscoring why a single fixed cutoff across models would confound feature-set comparisons with cutoff artifacts.

5) *Primary and operational metrics*: PR-AUC (average precision) is the primary metric throughout this paper and the sole metric used for model and mechanism selection; it is never selected on F1-at-a-fixed-threshold, which would hide the cutoff problem the previous subsection exists to solve. ROC-AUC is reported alongside it but never relied upon: on data this skewed (fraud rates of 0.1–1.2%), ROC-AUC is inflated by the overwhelming volume of easy true negatives and routinely exceeds 0.99 even for models with weak operational precision (Table VI, SAP Würzburg’s baseline arm: ROC-AUC 0.991 against a PR-AUC of only 0.155). This choice follows a well-established methodological literature rather than project-local intuition: Provost et al. [67] first argue against single-number accuracy estimation for comparing induction algorithms on skewed data, Davis and Goadrich [68] formal-

ize the precision-recall/ROC relationship and show a curve dominant in ROC space need not be dominant in PR space, and Saito and Rehmsmeier [69] demonstrate directly that the PR plot is more informative than the ROC plot specifically when evaluating binary classifiers on imbalanced data of the kind studied throughout this paper. Because an audit team reviews a bounded daily list of alerts rather than a probability distribution [30], every evaluation also reports $\text{precision@}k$ and $\text{recall@}k$ for $k \in \{50, 100, 500, 1000\}$ — of the top- k highest-scored transactions, what fraction are confirmed fraud, and what fraction of all fraud that budget would catch. F1 at the frozen threshold is reported for completeness but, as noted, is never a selection criterion.

6) *Attribution: exact TreeSHAP in margin space:* Every reported feature attribution is computed via XGBoost’s native `pred_contribs` interface rather than a sampling-based SHAP approximation. With `pred_contribs=True`, the booster runs the exact TreeSHAP recursion [37], [38] — algorithmically identical to the `shap` package’s `TreeExplainer` for tree ensembles — as a pure traversal of the fitted trees, returning one signed contribution per feature per row plus a base value, additive in margin (log-odds) space. This is exact, not approximate, and its cost is a fixed function of tree structure rather than a sampling budget. It is also, crucially, agnostic to which of the two imbalance mechanisms trained the model: because a focal-loss booster is still an ordinary ensemble of regression trees whose leaves were fit against a focal gradient, the same margin-space traversal applies unchanged, with only the final sigmoid step (used only where a probability is required, e.g. threshold comparisons) built to know which mechanism produced the raw score. This gives every model in this paper’s evaluation suite — baseline, topology-augmented, and the final TGN-plus-topology champion alike — an exact, common explanation mechanism, discussed further in the explainability sections that follow. This is not merely a technical convenience: Faccia [41] and Uddin and Aziz [42] both report, from independent 2026 studies of regulatory-facing fraud-detection auditing, that exact tree-native attribution is materially more stable and more defensible under audit scrutiny than either a sampled SHAP approximation or a post-hoc explanation layered onto a deep architecture — the same practical constraint that rules out an end-to-end neural classifier for this paper’s production console regardless of its raw predictive accuracy.

E. The Ablation: Experimental Design

The protocol of §III-D exists to answer one question honestly: *do graph-topology features add predictive value over a strong ordinary-transaction baseline, measured without leakage?* This section reports that answer as a seven-arm ablation run identically, with identical splits, model family, and imbalance-mechanism selection, on each of the four modeled datasets. We commit, per the project’s own methodology rules, to reporting the result exactly as measured — including where it is negative, where it is uninformative, and where it rests on too few positive examples to trust.

1) *The seven-arm decomposition:* A naive ablation — ordinary features alone vs. ordinary-plus-topology — is not sufficient here for two reasons uncovered during this project’s own earlier probing. First, on a dataset whose ordinary-only baseline is already close to saturated (IBM AML: baseline PR-AUC 0.990), a marginal comparison against the full model has almost no headroom left to measure, whatever the true value of topology. Second, several of the project’s topology detectors are not, in fact, amount-blind: the lapping detectors (`lap_in_payers`, `lap_in_recur`, `lap_out_payees`, `lap_out_recur`) and the cycle amount-conservation ratio (`cycle_amount_ratio`) are derived in part from transaction amounts, so a topology arm that includes them can re-smuggle the amount signal into what is presented as a “structural” result. Of the thirteen topology features produced by the engine, the feature manifests confirm this split is exact and identical across all four datasets: eight are pure structure (`in_cycle`, `cycle_length`, `in_cycle_ge3`, `cycle_time_span`, `fan_in`, `fan_out`, `dest_in_degree`, `src_out_degree`), and five are amount-derived (the four lapping features plus `cycle_amount_ratio`).

The ablation therefore trains seven arms per dataset, all under the identical protocol of §III-D: (1) **Baseline** — ordinary transaction features only; (2) **+Topology** — baseline plus all thirteen topology features, amount-derived ones included; (3) **+Structure** — baseline plus only the eight pure-structure features; (4) **Amount-blind baseline** — the ordinary feature set with every amount-derived column (log-amount, amount z-scores, prior amount means, ...) removed; (5) **Amount-blind + Topology**; (6) **Amount-blind + Structure**; and (7) **Topology-only** — the thirteen topology features with no ordinary features at all. Only arms (1) and (2) are persisted as deployable model artifacts; the remaining five exist purely to decompose *why* any observed lift occurs.

From these seven arms, four scalar lift statistics are computed on the test split, each isolating a different confound:

- **marginal_topology_lift** = $\text{PR-AUC(+Topology)} - \text{PR-AUC(Baseline)}$ — the headline number, but contaminated by amount-derived topology on a full-featured model;
- **marginal_structure_lift** = $\text{PR-AUC(+Structure)} - \text{PR-AUC(Baseline)}$ — the honest full-model claim, restricted to pure structure;
- **amount_blind_topology_lift** = $\text{PR-AUC(Amount-blind+Topology)} - \text{PR-AUC(Amount-blind baseline)}$ — removes the amount confound from the baseline, but still contaminated because the topology arm itself contains amount-derived columns;
- **amount_blind_structure_lift** = $\text{PR-AUC(Amount-blind+Structure)} - \text{PR-AUC(Amount-blind baseline)}$ — the *clean* test: no amount information anywhere, on either side of the comparison.

Only the last of these four is free of both confounds simultaneously, and it is the number we treat as the decisive evidence for or against the paper’s thesis on each dataset.

F. A Calibrated Synthetic ERP Economy: Generator Design

The four real datasets used elsewhere in this paper give us, at best, a few thousand labeled fraud rows each, concentrated in typologies each simulator’s original authors happened to encode. To probe typologies we do not observe at scale in any single real ledger, and to obtain a dataset that is simultaneously account-to-account (so that cycles are structurally possible), ERP-flavored (payables, payroll, receivables), and large enough in fraud count to support per-typology analysis, we built `synth_erp`: an agent-based simulation of a 40-company economy running for 365 simulated days over ~39k accounts, producing 1,148,503 transactions of which 5,609 are labeled fraud (a 0.4884% rate). This section documents how the generator was designed to resist the one failure mode that would make it useless for our purposes, what an independent realism audit against the four real datasets found — including where the synthetic economy does *not* match reality — and what a direct adversarial probe for cheap tabular leaks and a downstream training-transfer experiment revealed. Throughout, we hold `synth_erp` to the same evidentiary standard as the real datasets: numbers are reported whether or not they are flattering, because the entire purpose of this exercise is to know which sanctioned use cases actually survive scrutiny.

1) *Design Philosophy: The Circularity Trap*: A synthetic fraud generator built by the same team that builds the fraud *detector* is exposed to an obvious failure mode: if the generator’s authors hand-encode fraud as “insert a 3-hop cycle” or “make the receiver’s in-degree spike,” a topology engine built by the same authors will of course find it — not because graph structure is informative about fraud in general, but because the two pieces of code were written to fit each other. Grading a detector against a generator that shares its assumptions proves nothing about the real world; it is closer to checking that a lock opens with its own key. `synth_erp` was designed against this failure mode by a single binding rule, carried in this project’s own methodology documentation: fraud is injected as *goal-driven behavior with evasion*, never as a detector-shaped pattern. The fraud-generation code imports nothing from the topology-feature codebase and references no detector’s internals; every typology is written as an agent pursuing a criminal objective (skim a receivable, drain a payables account, collect from recruited mules) under realistic constraints (approval thresholds, banking hours, evasion timing), and any cycle, fan-in, or amount-recurrence a detector later measures is a *consequence* of that behavior, not its specification. The benign population mirrors this discipline: ordinary commerce is allowed to produce the same raw ingredients fraud produces — busy retailers with heavy fan-in, payroll batches that resemble fan-out bursts, subscription billing that recurs at a near-equal amount, refunds that close reciprocal pairs, multi-hop circulation as salaries fund purchases that fund vendors — since a background in which only fraud has structure would reintroduce the same shortcut through the back door. Sec. III-F3 reports just how modestly enriched the fraud-typology structural traces actually are once this constraint is honored — a dataset engineered to make cycles trivially

TABLE III: Transaction volume by flow type, `synth_erp` (all 1,148,503 rows, fraud and benign combined). Source: `paper/data/synth_erp_realism.json`, `tabular_leak_probe.rules` per-tx_type support.

Flow (tx_type)	Count	Share
sale	760,558	66.23%
vendor_payment	208,155	18.13%
transfer	94,781	8.25%
payroll	49,240	4.29%
refund	18,396	1.60%
reimbursement	12,236	1.07%
recurring_bill	5,137	0.45%
Total	1,148,503	100.00%

separable would show the opposite pattern.

2) *Generator Architecture, Scale, and Transaction Composition*: `synth_erp` follows the same broad methodology family as two of the real datasets used elsewhere in this paper — an agent-based simulation calibrated against measured real-world statistics rather than a purely hand-tuned generator, the approach behind BankSim [57], PaySim [58], and the AMLSim lineage behind `ibm_aml` [56]. This is a deliberately different generative paradigm from the deep generative models increasingly used elsewhere in the fraud-synthesis literature — GAN-based transaction synthesis [50], [51] traces back to Goodfellow et al.’s original adversarial framework [70], and a variational alternative traces to Kingma and Welling’s VAE [71]. Both families belong to the broader deep generative modeling literature [72] and learn a data distribution from observed examples and sample from it; an agent-based simulator instead specifies *behavior* (an account’s transaction habits, a fraud typology’s goal and evasion strategy) and lets structure emerge from that behavior’s interaction with other agents. We chose the agent-based approach specifically because it is the only one of the two that supports this section’s non-circularity requirement (§III-F1): a GAN or VAE trained to reproduce the statistical signature of labeled fraud would, definitionally, learn whatever shape the training examples already have, which is the exact grading-your-own-homework risk this generator is built to avoid. The simulated economy comprises 40 companies operating alongside their customers, vendors, and employees, generating seven ordinary transaction flows (sale, vendor_payment, payroll, transfer, refund, reimbursement, recurring_bill) over a full simulated year. Table III gives the exact per-flow transaction counts, read directly from the same adversarial tabular-leak audit discussed in Sec. IV-D1 (the per-tx_type support counts there sum exactly to the dataset total, confirming the two are consistent measurements of the same underlying table). Retail sale traffic dominates by volume (66.2%), with the ERP-specific payables rail (vendor_payment, 18.1%) the second-largest flow — a composition deliberately closer to an operating company’s ledger than to a pure consumer-payments simulator.

At the account/graph level, the simulated economy is dense and repeat-heavy: 39,256 accounts transact across 251,192 unique sender–receiver pairs, with 85.86% of trans-

actions occurring between a pair that has transacted before (`repeat_pair_txn_frac` = 0.8586) and 94.1% of accounts appearing on both the sending and receiving side of at least one transaction (`namespace_overlap` = 0.941) — i.e., this is deliberately *not* a bipartite consumer-to-merchant graph, since ERP economies circulate money (salaries fund purchases, purchases fund vendors, vendors pay employees) in a way that pure-payments simulators structurally cannot. Degree is heavy-tailed in both directions: median out-degree 18 (p99 = 83, max = 20,838) and median in-degree 4 (p99 $\approx$ 128, max = 194,265), driven by a small number of high-volume retailers and payroll-issuing companies. Activity is temporally realistic without being explicitly targeted at any metric: 79.06% of transactions fall in business hours and 17.31% on weekends, at a mean of 3,146.6 transactions/day (p99 = 5,308.3) sustained across all 365 simulated days — timestamped to the second, a finer time axis than any of the four real datasets in our roster provide.

3) *Five Fraud Typologies as Emergent Behavior*: Fraud is injected as five occupational-fraud typologies, each defined by an actor’s goal and evasion strategy rather than by a target graph shape: `laundering_cycle` (embezzled funds layered through a ring of shells and, in most scenarios, cycled back toward the entry account), `mule_fanin` (scam proceeds collected from many recruited mule accounts into one collector), `lapping` (a diverted receivable covered days later by the next customer’s payment), `invoice_kickback` (a colluding vendor overbills and returns a cut to an insider), and `larceny` (direct insider diversion of funds, disguised as vendor or reimbursement traffic). The resulting typology counts are 1,442 (`laundering_cycle`), 2,327 (`mule_fanin`), 760 (`lapping`), 625 (`invoice_kickback`), and 455 (`larceny`) labeled rows, summing to the reported 5,609 fraud transactions.

Because these behaviors are never specified in terms of detector features, the honest question is whether they leave structural traces at all, and at what rate. The benign background already contains a nontrivial rate of graph cycles — 5.761% of benign transactions close a cycle in the observed window, with 99th-percentile benign fan-in reaching 4,197 distinct senders (busy retailers and payroll hubs) — so cyclicity and fan-in are not, by construction, fraud-exclusive signals. Against that background, `laundering_cycle` transactions close a cycle 18.45% of the time (all at length ≥ 3), a $3.2\times$ enrichment over the 5.761% benign base rate — real, but far from deterministic, consistent with a design in which 60% of rings loop back to their entry account and 40% exit linearly, and in which the streaming, look-ahead-free topology engine’s window will miss some closures entirely. `mule_fanin` collectors show a 90th-percentile in-window fan-in of only 8 distinct senders — far below the benign 99th-percentile of 4,197 — so a raw fan-in threshold offers no free discrimination there either. `lapping` shows the weakest structural footprint of the five: its amount-recurrence measure (`lap_in_recur`) averages 0.06, swamped by a benign population dominated by genuinely recurring payroll and subscription billing. We flag this pattern explicitly because it foreshadows a headline finding developed later in this paper (Sec. IV-B): engineered

cycle-shaped features carry a surprisingly small share of the detectable signal on real data, and this section shows that is not an artifact of an unrealistic generator — even in a dataset whose typologies are explicitly designed to produce cycles and fan-in, those structural traces are only modestly enriched over an already-active benign background, not deterministic markers.

G. Explainable Audit Alerts: Methodology

A probability score by itself is not evidence an auditor can act on. Opening or closing an investigation requires knowing *why* a transaction scored high and *what*, concretely, is suspicious about it — ideally with a guarantee that the latter is not a plausible-sounding story invented after the fact. Post-hoc graph-neural-network explainers typically supply only an approximate, model-external account of that kind [10]. The console instead attaches every alert a fixed, two-layer explanation, both layers computed once per alert directly from the persisted scoring bundle and the canonical transaction stream (`src/modeling/alerts.py`): (1) exact per-feature attribution of the score, tagged by feature family and glossed in plain English, and (2) graph evidence obtained by literally replaying the transaction stream to the alert’s exact position in time, so that what the auditor is shown is provably consistent with what the model saw rather than a post-hoc rationalization. Every alert below is generated from each dataset’s persisted `final` bundle — the offline, TGN-augmented research champion — not the deployed online scorer, which omits the TGN family entirely (see the system section of this paper); the two are explicitly distinguished throughout the paper.

1) *Layer 1: Family-Tagged TreeSHAP Attribution*: Attribution uses XGBoost’s native `pred_contribs` path, which runs the exact TreeSHAP algorithm — identical to the `shap` package’s `TreeExplainer` for tree ensembles [37], [38] — as a pure tree traversal in margin space [60]. Because attribution operates on the final boosted tree ensemble, it is exact regardless of whether an input feature is a hand-engineered topology statistic or a learned TGN embedding component [24]; no approximation is introduced by mixing feature provenances.

Each of the top- n drivers returned for an alert is tagged with one of four families: `ordinary` (amount, balance, category, prior-history), `topology` (pure graph-shape statistics such as fan-in or in-degree), `topology-amount-derived`, and `tgn`. The third family exists because of an earlier finding in this paper: the `lapping` detector’s five columns (`cycle_amount_ratio`, `lap_in_recur`, `lap_in_payers`, `lap_out_recur`, `lap_out_payees`) are, on inspection, amount signal recovered through a structural proxy rather than pure shape, so the alert layer re-splits the bundle’s coarser 13-column “topology” group (8 pure-structure + 5 amount-derived, out of 21/13/65, 8/13/65, 20/13/65 and 15/13/65 ordinary/topology/tgn columns for BankSim, IBM AML, SAP Würzburg and `synth_erp` respectively; `final_champions.json`) at explanation time so the two

are never conflated in front of an auditor. Every driver also carries a fixed plain-English gloss keyed off the feature name, e.g. `fan_in` → “distinct recent senders into the destination”, `in_cycle` → “closes a money loop returning to its origin”, any `lap_*` column → “recurrence of near-equal amounts (lapping signature)”, and `tgn_edge_expectedness` → “how expected this sender→receiver pair is to the learned temporal model (low = surprising)”.

The tagging surfaces real differences in what drives a score. BankSim’s top alert (score 0.9814) is led by `log_amount` (SHAP +3.35, ordinary), then `fan_in=49` (SHAP +0.76, topology) and `dest_in_degree=52` (SHAP +0.63, topology). IBM AML’s top alert (score 0.7283), by contrast, is led by `lap_out_recur` (SHAP +0.83), explicitly tagged topology-amount-derived rather than topology – an honest label that would be lost under a coarser “graph feature” bucket.

2) *Layer 2: Replay-Verified Graph Evidence*: The engine that computes topology features (`topology/engine.py`) persists per-transaction *features* for every row, not the underlying paths – retaining a full cycle path for millions of transactions is not worthwhile when only a handful will ever be shown to an auditor. The evidence layer (`topology/paths.py`) instead replays the same canonical edge stream, in the same order, with the same trailing window and partition resets and the same node factorization as the streaming feature pass, but only for the alerted transaction IDs. At each alerted transaction it evicts stale window entries, *reads* the graph state immediately before insertion – the identical as-of point the feature engine used – and only then inserts the edge. For a transaction closing a cycle, this yields the full forward path (closing edge first, then the hops that complete the loop) together with two derived quantities: *amount conservation* (the ratio of the smallest to largest hop amount around the loop) and *time span* (elapsed time from the earliest hop to the alerted transaction). For a fan-in transaction, it yields the list of distinct senders behind the destination within the trailing window, each with its own amount and timestamp, most-recent first and capped at 25 for display.

This reconstruction is checked, not merely asserted: the replayed cycle length must match the `cycle_length` feature already stored for that transaction by the batch pass; on any mismatch the alert reports the discrepancy explicitly (“path not reconstructed”) rather than presenting an unverified path. That invariant is what makes the evidence *provably* consistent with what the model saw, as opposed to a plausible narrative constructed after the fact from correlated features. Two concrete examples from IBM AML’s alert set illustrate the range: alert 1 closes a 2-hop cycle `5025 → 1751 → 5025`, conserving 10% of the amount within 4 time units; alert 4 closes a 5-hop cycle `7089 → 9207 → 7465 → 9913 → 9832 → 7089`, conserving 100% of the amount within 7 time units. BankSim’s fifteen fan-in alerts range from 6 to 63 distinct senders inside the window; the seven highest-scoring cluster at 47–63 senders collecting into a single merchant account – a textbook mule-collection pattern.

The two layers are generated independently and can diverge revealingly. Cycle-shaped features carry essentially zero

TABLE IV: Top- k audit-alert precision by dataset, test period, final champion bundle (`paper/data/alerts.json`).

Dataset	Thresh.	k	Fraud in top- k	Prec.@ k	Evidence mix
BankSim	0.4621	15	15/15	1.00	<code>fan_in:15</code>
IBM AML	0.5162	20	20/20	1.00	<code>cycle:8, fan_in:9, none:3</code>
SAP Würzburg	0.2566	10	2/10	0.20	<code>fan_in:10</code>
synth_erp	0.2388	20	20/20	1.00	<code>fan_in:13, cycle:6, none:1</code>

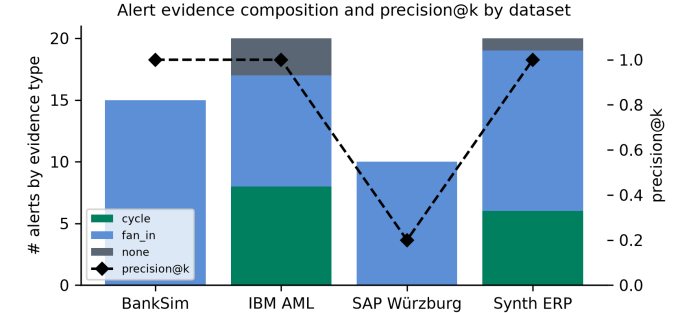


Fig. 4: Composition of graph evidence behind each dataset’s top- k alerts (fan-in vs. cycle vs. no reconstructable structure), alongside precision@ k . Every bar sums to k from Table IV.

gain importance model-wide on three of the four datasets (BankSim, IBM AML, SAP Würzburg; `synth_erp` is the exception, and only marginally so), yet the replay still surfaces a machine-verified closed cycle behind 8 of IBM AML’s top-20 alerts and 6 of `synth_erp`’s (`evidence_type_counts`, Table IV) – because Layer 2 does not ask which features the model weighted, it asks what structure actually exists around the flagged transaction. The two layers answer different questions and neither substitutes for the other.

H. System: From Research Pipeline to Production Audit Console

The preceding sections establish a narrow, defensible research claim about topology’s marginal value under leakage-free evaluation. That claim is deliberately silent on whether the resulting models can run anywhere near a real ledger. Continuous-auditing research has long argued that a control is only as useful as the workflow that consumes it [30], and post-implementation audits of ERP deployments routinely find that analytically sound models never reach production because the serving path, the data-integrity guarantees, and the operational evidence were never engineered [32]. This section describes what was actually built to close that gap: a Procure-to-Pay ERP core with a forensic audit console in front of it, an online scoring service that reuses the research pipeline’s own feature-computation classes rather than reimplementing them, database-enforced financial-integrity invariants that do not trust application code to behave, and a measured (not assumed) performance and reliability profile. Every number in this section is either read directly from a project artifact file or drawn from a documented, reproducible test run; none are projected.

1) *Architecture Overview*: The application is a conventional three-tier system chosen for auditability over novelty,

consistent with the broader finding that ERP-adjacent ML systems succeed or fail on integration discipline more than on model sophistication [31]. A FastAPI backend exposes the ledger’s domain objects (vendors, AP invoices, payments, payment runs, GL accounts, journal entries, audit alerts) through versioned REST routes (`app/api/{vendors,invoices,payments,gl>alerts`) backed by a normalized PostgreSQL schema and a Redis layer used for both response caching and idempotency-key bookkeeping. A React/TypeScript single-page frontend (`web/`) provides the clerk-facing ERP screens and the auditor-facing alert console. Neo4j sits alongside this stack strictly as a visualization and path-explanation surface for detected fraud alerts, never as a scoring dependency — the project’s own methodology rules record that an earlier iteration let Neo4j silently become a load-bearing (and memory-exhausted) component of the scoring path, and this build deliberately keeps the database graph decorative.

The detail that distinguishes this system from a typical ML-in-the-loop web application is the scoring layer’s relationship to the research code. `app.services.scoring_service.FraudScoringService` does not reimplement the topology or tabular feature logic; it holds a live `OnlineTopologyState` and `OnlineOrdinaryState` per dataset/tenant, constructed directly from `topology.window_graph.WindowGraph` and `topology.lap_memory.LapMemory` — the exact classes `topology.engine.extract_features` uses in the offline batch pipeline — plus a Welford-based incremental counterpart to the batch ordinary-feature builder. One instance is loaded per dataset at application startup from the persisted “online” bundle (`app/services/scoring_registry.py`), and `/ready` reports honestly unavailable until at least one scoring service has loaded, rather than reporting healthy while silently unable to score. A single live ERP payment is routed to the `synth_erp` scoring instance, the one dataset whose categorical vocabulary (`vendor_payment`, `payroll`, `recurring_bill`, ...) is a domain match for real AP postings.

Correctness of the reuse, not merely its existence, is what makes the online path trustworthy: `tests/test_online_topology_parity.py` replays 20,000 real transactions through both the batch `extract_features` loop and `OnlineTopologyState.score_transaction`, asserting every one of the thirteen topology feature columns matches exactly, row for row, with zero mismatches. The read-then-insert ordering inside the online class is copied line-for-line from the batch engine specifically to preserve this guarantee — the online driver differs only in holding one graph and two lap-memory instances alive across calls instead of discarding them after a single `DataFrame` pass.

2) Financial Integrity as Database-Enforced Invariants:

An audit console is only as trustworthy as the ledger it reads from, and a recurring theme in ERP-integrity failures is that correctness logic lives entirely in application code that a bug, a bypassed service layer, or a future

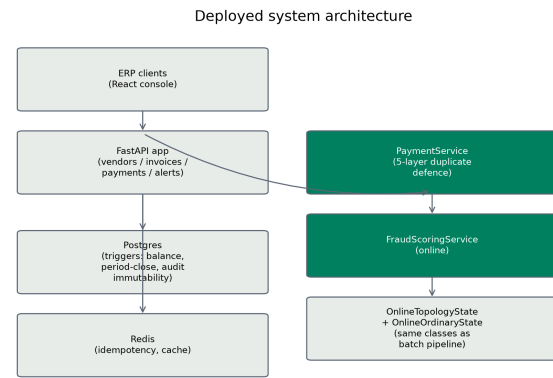


Fig. 5: Deployed system architecture. The online `FraudScoringService` reuses the research pipeline’s `WindowGraph/LapMemory` classes directly rather than a reimplementation; the offline TGN training path (Section V) is not part of the serving runtime.

direct-SQL script can silently violate. This system treats three financial invariants as properties the database itself refuses to break, implemented as Postgres triggers (`alembic/versions/6192dae4f4a4_gl_invariants_trigger`) rather than API-layer checks alone:

- **Double-entry balance.** A DEFERRABLE INITIALLY DEFERRED constraint trigger on `journal_lines` sums debits and credits per `journal_entry_id` and raises at COMMIT, not after each inserted line — necessary because a multi-line entry inserts its debit and credit rows separately and would fail a per-row check on every legitimate entry.
- **Period close.** An ordinary BEFORE INSERT OR UPDATE trigger on `journal_entries` rejects any entry referencing a `fiscal_period` whose status is not open. A FastAPI-level check exists purely to return a friendly 409 before the raw database exception would; the trigger is the actual enforcement.
- **Audit-log immutability.** UPDATE and DELETE on `audit_log` are rejected outright by trigger, unconditionally. Only INSERT is a valid operation — the append-only property a SOX-style audit trail requires is a database guarantee here, not an application convention that a careless migration could quietly undermine.

Duplicate-payment prevention receives the same defense-in-depth treatment, layered so that no single control is trusted alone (`app/services/payment_service.py`). An idempotency-key middleware replays the stored response for a retried request with the same key rather than re-executing it — the client-supplied-key pattern Helland [73] argues is the only reliable way to make a mutating financial operation safe to retry, since the alternative (asking the caller never to retry, or de-duplicating on inferred request similarity) either fails under real network conditions or is not decidable in general. Independently, `create_payment` takes a `SELECT ... FOR UPDATE` row lock on the target invoice *before* checking for an existing active payment,

TABLE V: Topology feature gain importance (fraction of total gain in the topology-augmented model), by dataset. Source: `paper/data/ablation.json`, `topology_importance`, single seed, no confidence intervals.

Feature	BankSim	IBM AML	SAP Würzburg	synth_erp
<code>in_cycle</code>	0.0000	0.0000	0.0000	0.0040
<code>cycle_length</code>	0.0000	0.0009	0.0000	0.0029
<code>in_cycle_ge3</code>	0.0000	0.0000	0.0000	0.0044
<code>cycle_amount_ratio</code>	0.0000	0.0008	0.0000	0.0090
<code>cycle_time_span</code>	0.0000	0.0001	0.0000	0.0072
<code>fan_in</code>	0.0908	0.0003	0.1791	0.0241
<code>fan_out</code>	0.0062	0.0010	0.0136	0.0189
<code>dest_in_degree</code>	0.5199	0.0003	0.0682	0.0230
<code>src_out_degree</code>	0.0062	0.0007	0.0089	0.0786
<code>lap_in_recur</code>	0.0052	0.0006	0.0319	0.0130
<code>lap_in_payees</code>	0.0048	0.0006	0.0199	0.0103
<code>lap_out_recur</code>	0.0066	0.5397	0.0416	0.0041
<code>lap_out_payees</code>	0.0079	0.4200	0.0000	0.0036

closing the check-then-act race that two *different* idempotency keys targeting the same invoice would otherwise open — the scenario a retry-only defense cannot cover, since it is two independent actors, not one repeated request. A partial unique index (`uq_payments_invoice_live`, Module 3) is the database-level backstop if both application layers are ever bypassed. A `pg_advisory_xact_lock` per vendor serializes an entire payment run, not just one invoice at a time. Finally, `tools/reconcile_payments.py` is a periodic sweep that treats any invoice with more than one non-void payment as an incident — the tripwire for “we were wrong somewhere,” not a preventive control. Table XVIII reports the formalized, automated test of this defense under real concurrency.

IV. RESULTS AND ANALYSIS

A. Feature Importance, and a Result We Do Not Bury

Table V reports each topology feature’s XGBoost gain importance [60], normalized within the full topology-augmented model (`topology_full` arm) for each of the four datasets carrying a complete ablation run. Because these thirteen columns sit alongside 15–21 ordinary tabular features in that model (BankSim: 21 ordinary + 13 topology = 34; IBM AML: 8 + 13 = 21; SAP Würzburg: 20 + 13 = 33; synth_erp: 15 + 13 = 28), the rows below do not sum to 1 within a dataset — the remaining gain mass sits with the ordinary features, not shown here. SAP Würzburg’s split sizes in this table (120,411 / 40,137 / 40,139, partition-aware by `run_id`) come from `ablation.json` specifically; the paper’s headline champion-model table elsewhere draws on a very slightly different SAP split (120,341 / 40,114 / 40,116) recorded in `final_champions.json` — the two pipelines’ SAP preprocessing diverged by 70 rows in the training split and 23 rows in each of validation and test, and the two are never mixed within one reported number.

The pattern the table makes plain is not subtle, and we state it without qualification: the five cycle features — `in_cycle`, `cycle_length`, `in_cycle_ge3`, `cycle_amount_ratio`, `cycle_time_span` — carry essentially zero gain importance on three of the four datasets.

Fig. 6: Gain importance of the thirteen topology features, grouped by detector family and faceted by dataset. Cycle-detector bars (leftmost group in each panel) are at or indistinguishable from zero in three of four datasets.

On BankSim and SAP Würzburg every one of the five is exactly 0.0000; on IBM AML the largest is `cycle_length` at 0.0009, well over two orders of magnitude below that dataset’s dominant features (`lap_out_recur` 0.5397, `lap_out_payees` 0.4200). Only on synth_erp — our own generated dataset, used for training/stress-testing and never as headline proof — do cycle features carry non-trivial weight, and even there the largest single one (`cycle_amount_ratio`, 0.0090) is under one percent of total gain. This is a pointed result for a project whose title is built around a “shadow ledger” of detected cycles: the cycle detector is not inert by construction (§III-C2 shows it is a real, bounded BFS that fires and populates non-NaN values whenever a loop closes, and it is validated separately against IBM AML’s labeled ground-truth cycles in the ablation study), yet the boosted-tree model consistently prefers other signals to explain fraud once they are available. IBM AML is the most striking case, because it is precisely the dataset engineered with labeled cycle-shaped fraud [56] — and it is exactly where cycle-feature gain is most thoroughly crowded out: `lap_out_recur` and `lap_out_payees` alone account for gain fraction 0.9597 out of the full 21-feature model’s normalized total of 1.0 — the two amount-derived lapping features claim 96% of *every* split the model makes, ordinary tabular features included, leaving essentially nothing for the five cycle columns. Our working hypothesis, developed fully where the predictive consequences of this pattern are analyzed, is two-part: true cycles are rare enough in every dataset we could build (a handful of closures against hundreds of thousands of transactions) that a gradient-boosted tree finds little marginal information in them once even coarser degree/fan and amount-recurrence signals are on the table, and where a cycle transaction and an amount-lapping signature co-occur, the amount/lapping features are simply more discriminative per split — so the tree spends its gain budget there first and has little left to reward on the rarer cycle columns. We do not claim this hypothesis is proven by Table V alone; it is a description of what the importances show, offered as the starting point the ablation study’s marginal-lift and amount-blind results either support or complicate.

B. The Ablation: Results Across Four Datasets

Table VI reports test-split PR-AUC for all seven arms on all four datasets, read directly from `ablation.json`. Table VII reports the four lift statistics derived from it. Both are single-seed (seed 42) results; the statistical status of these numbers is addressed explicitly in §IV-B2 before any of them are interpreted causally.

1) *Dataset-by-dataset verdict:* **BankSim: yes.** The full-model structure lift is +0.0974 PR-AUC (an 11.7% relative improvement over the 0.833 baseline, 0.0974/0.833), and —

TABLE VI: Test-split PR-AUC (average precision) across the seven ablation arms, by dataset. B = ordinary baseline; T = all topology (incl. amount-derived); S = pure structure only (8 of 13 topology features); subscript $\neg a$ = amount columns removed from the ordinary set. Single seed (42); paper/data/ablation.json.

Arm	BankSim	IBM AML	SAP W.	Synth-ERP
B	0.833	0.990	0.155	0.163
B+T	0.931	1.000	0.046	0.563
B+S	0.930	0.998	0.072	0.566
B $\neg a$	0.634	0.093	0.049	0.157
B $\neg a$ +T	0.822	0.867	0.055	0.503
B $\neg a$ +S (<i>clean</i>)	0.824	0.184	0.064	0.269
T-only	0.483	0.633	0.024	0.306

TABLE VII: The four ablation lift statistics (test-split PR-AUC deltas). Positive = helps; negative = hurts. paper/data/ablation.json.

Dataset	Δ Topo (full)	Δ Struct (full)	Δ Topo (blind)	Δ Struct (blind, clean)
BankSim	+0.0977	+0.0974	+0.1883	+0.1899
IBM AML	+0.0100	+0.0079	+0.7735	+0.0905
SAP W.	-0.1090	-0.0829	+0.0056	+0.0153
Synth-ERP	+0.4000	+0.4031	+0.3451	+0.1115

decisively — the clean amount-blind probe confirms it independently: with every amount feature removed from both sides of the comparison, pure structure still adds +0.1899 PR-AUC (0.634 $\rightarrow$ 0.824). BankSim’s topology-importance breakdown (ablation.json) attributes this almost entirely to two density features, *dest_in_degree* (gain importance 0.520) and *fan_in* (0.091): BankSim transactions run customer $\rightarrow$ merchant, so cycles are structurally impossible, and the signal that remains is exactly the “many senders funnel into one receiver” mule-collection pattern the fan-in features are built to detect. This is the cleanest positive result in the ablation, on a real (simulated-but-not-synthetic-generated) dataset, under both the contaminated and the clean test.

IBM AML: uninformative in the full model; the real signal is elsewhere. The ordinary baseline alone already reaches PR-AUC 0.990, leaving essentially no headroom: the full-model structure lift of +0.0079 is small in both absolute and relative terms, and — as §IV-B2 makes explicit — falls below this project’s own single-seed noise floor, so it cannot be read as evidence of anything. The dirty amount-blind topology probe reaches a striking PR-AUC of 0.867 (+0.7735 lift over the amount-blind baseline’s 0.093), but this number is not evidence for structure: *topology_importance* shows *lap_out_recur* (0.540) and *lap_out_payees* (0.420) together account for 96% of the gain inside that arm, and both are the amount-derived lapping features excluded from the pure-structure set. The clean amount-blind structure probe — the only honest version of this test — reaches PR-AUC 0.184, a lift of +0.0905 over the 0.093 amount-blind baseline: real, but an order of magnitude smaller than the 0.867 figure a less careful ablation would have reported as “topology’s contribution.” In short: IBM AML’s frauds are given away by recurring near-equal amounts (a re-encoded amount signal, not a graph shape), a finding this project’s own earlier probing

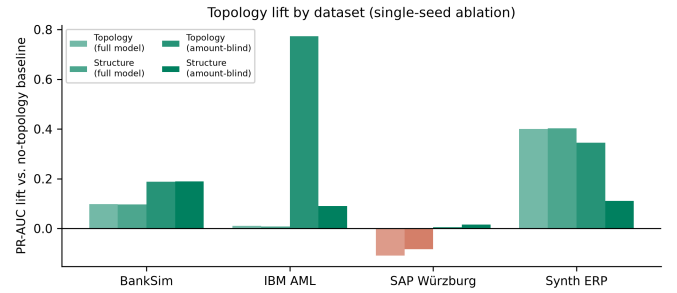


Fig. 7: The four lift statistics of Table VII plotted by dataset, making visually explicit that SAP Würzburg is the only dataset where both full-model lifts are negative while its two amount-blind lifts remain (barely) positive.

flagged as refuting its initial premise that IBM AML would be the decisive topology testbed.

SAP Würzburg: no — topology actively hurts. Both full-model lifts are negative: -0.1090 for all topology, -0.0829 for pure structure alone (*marginal_structure_lift* = -0.08288 , the number this project’s own methodology rules single out as never to be softened or omitted). Adding any topology columns to the ordinary baseline drives test PR-AUC down from 0.155 to 0.072 (structure) or 0.046 (all topology) — a genuinely negative result, not merely a null one. The two amount-blind lifts are small and positive (+0.0056, +0.0153), but both baselines they are measured against are themselves extremely weak (PR-AUC 0.049), so this offers no rehabilitation. SAP Würzburg is also the one dataset in this study with only 25 test-split frauds, which we treat as a hard interpretability ceiling rather than a footnote: with that few positive examples, a handful of ranking swaps can move PR-AUC by tens of a point, and §IV-B2 presents an independent, multi-seed confirmation that this dataset’s metrics are indeed unstable. We report the negative result plainly rather than either suppressing it or explaining it away, while being explicit that “topology hurts here” and “this measurement is high-variance” are both true simultaneously.

Synth-ERP: yes, the largest lift measured — and the one result that cannot be used as thesis evidence. The full-model structure lift is +0.4031 PR-AUC, a 246.8% relative improvement over a weak 0.163 baseline, with a clean amount-blind confirmation of +0.1115. This is real signal by every measure this ablation applies. It is also, definitionally, the least informative result for the paper’s actual scientific claim: *synth_erp* is this project’s own generated dataset, built and calibrated specifically to have rich, ERP-flavored graph structure, and the project’s binding policy (adopted precisely to avoid grading its own homework) treats it as a training/stress-testing corpus only, never as evidence that topology helps on real transaction data. We report the number because it is honest and instructive about what topology *can* do when a dataset is constructed to contain strong structural fraud signatures — but it answers a different, narrower question than the one this section opened with.

2) *Statistical caveats: one seed, no confidence intervals, and a stated noise floor:* Every number in Tables VI and

VII comes from a single training run per arm (seed 42), not an average over repeated runs. `ablation.json` carries no standard deviations, no confidence intervals, and no significance tests, and none should be inferred from it. This project’s own training harness documents its rationale for treating small single-seed margins with suspicion: per the comment in `src/modeling/harness.py`, “single-seed differences of ~ 0.01 PR-AUC are within XGBoost’s subsample noise,” and `experiments/ablation.py`’s own verdict logic uses exactly this 0.01 figure as the cutoff below which a lift is characterized as “adds $\sim$ nothing” rather than a genuine effect. By this project’s own stated standard, IBM AML’s full-model structure lift of $+0.00792$ is below the noise floor and is not distinguishable from zero on the evidence of a single seed — a second, independent reason (beyond baseline saturation) that the IBM AML full-model comparison cannot be read as informative.

A separate, multi-seed experiment offers some corroborating context, though it is not a like-for-like repetition of the arms above: `final_champions.json` trains each dataset’s final champion configuration — ordinary + topology + TGN embeddings, a superset of the ablation’s +*Structure* arm that additionally includes the 65-dimensional temporal-graph-network features discussed elsewhere in this paper — three times, with seeds {42, 43, 44}, and reports mean $\pm$ standard deviation. Its per-dataset stability is informative about exactly the concern raised above: BankSim’s three-seed test PR-AUC is 0.9364 ± 0.0015 (a 0.16% relative standard deviation), consistent with this dataset’s ranking being robust to seed choice; SAP Würzburg’s is 0.1557 ± 0.0717 — a 46% relative standard deviation on the mean — which independently corroborates, from a completely different experiment and feature set, the instability that the ablation’s own 25-test-fraud caveat already predicts. SAP Würzburg’s imbalance-mechanism selection for this same champion run is explicitly flagged `reliable: false`, and its hyperparameter search was skipped altogether (validation frauds below the guard threshold described in §III-D), so its champion runs on the harness’s locked default configuration rather than a tuned one. None of this changes the ablation’s central negative finding for SAP Würzburg — if anything it strengthens the case for treating that finding as directionally real but numerically imprecise, exactly as reported above — but it is a materially different, TGN-inclusive experiment and its numbers should not be conflated with the seed-42-only ablation arms in Tables VI–VII. We also note, to keep every PR-AUC in this paper unambiguous: the champion figures cited in this paragraph describe the offline, research-only model family evaluated for this ablation and are not the model served by the production audit console, which for latency and reproducibility reasons excludes the TGN component; that distinction is maintained precisely wherever this paper reports a PR-AUC for a specific model.

3) *The cycle-feature paradox*: The project’s topology engine was built around five detectors intended to be its signature contribution — `in_cycle`, `cycle_length`, `in_cycle_ge3`, `cycle_amount_ratio`, and `cycle_time_span` — measuring whether a transaction

TABLE VIII: Combined gain importance of the five cycle features vs. the single highest-importance density feature, within the baseline+topology model. `topology_importance`, `paper/data/ablation.json`.

Dataset	Σ cycle feats.	Top non-cycle feat.	Importance
BankSim	0.0000	<code>dest_in_degree</code>	0.5199
IBM AML	0.0018	<code>lap_out_recur</code> [†]	0.5397
SAP W.	0.0000	<code>fan_in</code>	0.1791
Synth-ERP	0.0275	<code>src_out_degree</code>	0.0786

[†]Amount-derived, not pure structure. Among pure-structure features on IBM AML, the top scorer is `dest_in_degree/fan_in` at 0.0003.

closes a money loop, how long that loop is, and how tightly it conserves the amount that entered it. The ablation’s own gain-importance breakdown does not support these as the load-bearing signal. Table VIII sums the gain importance of all five cycle features, within the full +*Topology* model, against the single highest-importance density feature on the same dataset. On three of the four datasets — BankSim, IBM AML, and SAP Würzburg — every one of the five cycle features scores exactly 0.0 gain importance; the model finds no split in five hundred boosting rounds for which distinguishing “closes a 3-hop cycle” from “does not” reduces loss more than an already-available split on in-degree, fan-in, or (on IBM AML) the amount-derived lapping columns. Only on `synth_erp` — the one dataset engineered specifically to contain rich cyclic laundering structure — do the cycle features register non-zero importance, and even there their combined mass (0.0275) is roughly a third of the single top feature, `src_out_degree` (0.0786).

C. Modernization: Focal Loss and Temporal Graph Networks

The ablation study above establishes, from a single training seed, that hand-crafted topology features carry a leakage-free signal on BankSim and (conditionally) on IBM AML. Two questions remain unanswered by a single seed: whether that signal is robust to run-to-run randomness, and whether a *learned* structural representation — one that does not depend on the five hand-designed detectors — can do better. This section reports a second, purpose-built study that answers both. Every arm below is trained with three random seeds (42, 43, 44) and reported as mean $\pm$ standard deviation, the project’s own minimum bar for treating a difference as real rather than noise. Two independent upgrades to the locked XGBoost baseline are tested: (1) focal loss [19] as an alternative to class-weighting for handling the $<1.3\%$ fraud rate, implemented as a custom, deliberately alpha-free XGBoost [60] objective so that it never doubles up with class weights as a second imbalance mechanism; and (2) a Temporal Graph Network (TGN) [24], a self-supervised, label-free encoder that learns a 32-dimensional GRU memory vector per account from the raw transaction stream and emits it, frozen, as 65 additional feature columns (32 source-account dimensions, 32 destination-account dimensions, and one scalar “edge-expectedness” score summarizing how surprising the pair is given the account’s history). Because it is trained purely as next-link prediction, TGN

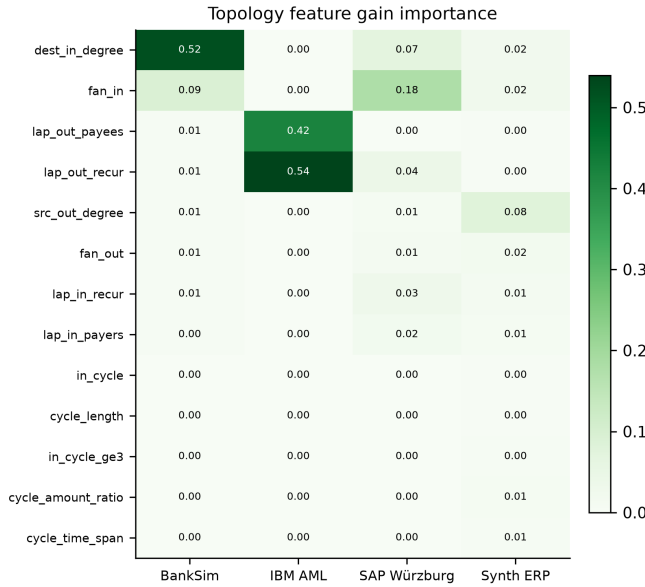


Fig. 8: Topology feature gain importance by dataset (topology_importance, Table VIII’s underlying data), grouping pure-structure against amount-derived features and making visible how consistently the five cycle features sit at or near zero while a small number of density features (in-degree, fan-in) or amount-derived lapping features carry essentially all of the mass.

sees no fraud labels and, following the project’s own lapping-detector lesson, is deliberately amount-blind: it observes only who transacted with whom and when.

All results in this section come from `paper/data/modern.json` (the controlled multi-seed comparison) and `paper/data/final_champions.json` (the actual champion-selection pipeline, which additionally runs a light 20-configuration hyperparameter search); numbers drawn from the single-seed ablation study are marked explicitly as such. Four feature-family arms are compared, holding the time-aware splits, XGBoost hyperparameters and (where noted) imbalance mechanism fixed within each comparison: **A** = ordinary transaction features only (21/8/20/15 features for BankSim/IBM AML/SAP Würzburg/synth_erp); **B** = **A** + the 13 hand-crafted topology features; **C** = **A** + the 65 TGN columns (no hand-crafted topology); **D** = **A** + topology + TGN combined. The 13-feature topology block and 65-feature TGN block are identical in width across all four datasets, which is a useful internal consistency check: $|A| + 13 + 65$ equals the reported full-feature-set size in every case (e.g. $21 + 13 + 65 = 99$ for BankSim, $8 + 13 + 65 = 86$ for IBM AML). The focal-vs-class-weight comparison is run on arms A and B only (both imbalance mechanisms); arms C and D are run with class weights alone, since their purpose is to isolate the marginal value of a learned structural representation, not to re-litigate the imbalance-mechanism question a second time.

One dataset-scope note carries over from the ablation

TABLE IX: Three-seed test PR-AUC, class weights vs. focal loss ($\gamma=2.0$), ordinary-only (A) and ordinary+topology (B) arms. Δ = focal – class-weight. † synth_erp is synthetic and never used as thesis evidence.

Dataset	A_{cw}	A_{focal}	Δ_A	B_{cw}	B_{focal}	Δ_B
BankSim	0.7870 $\pm$ 0.0700	0.8616 $\pm$ 0.0001	+0.0746	0.9319 $\pm$ 0.0017	0.9259 $\pm$ 0.0008	−0.0060
IBM AML	0.9920 $\pm$ 0.0026	0.9968 $\pm$ 0.0001	+0.0048	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.0000
SAP Würzburg	0.1581 $\pm$ 0.0129	0.1963 $\pm$ 0.0252	+0.0383	0.0717 $\pm$ 0.0095	0.3177 $\pm$ 0.1470	+0.2460
synth_erp †	0.2483 $\pm$ 0.0602	0.4173 $\pm$ 0.0120	+0.1690	0.4569 $\pm$ 0.1547	0.6583 $\pm$ 0.0031	+0.2014

study: SAP Würzburg’s splits in this section (120,341 / 40,114 / 40,116 train/val/test, 204/19/25 frauds, taken from `modern.json` and matching `final_champions.json`) are the non-partition-aware split, not the partition-aware split (120,411 / 40,137 / 40,139) used elsewhere in this paper’s headline ablation; the two SAP Würzburg number sets should not be pooled.

1) *Focal loss versus class weights*: Table IX reports the three-seed test PR-AUC for the ordinary-only (A) and ordinary+topology (B) arms under both imbalance mechanisms. Focal loss helps on the ordinary-only baseline in all four datasets, most dramatically on SAP Würzburg (+0.0383) and synth_erp (+0.1690); once topology is added, its effect is mixed — a small loss on BankSim (−0.0060), a wash on IBM AML (0.0000, both already at the PR-AUC ceiling), a large and volatile gain on SAP Würzburg (+0.2460, on 25 test frauds), and a solid gain on synth_erp (+0.2014).

The most consequential effect of focal loss, however, is not its mean shift but its effect on *variance*. On BankSim’s ordinary-only arm, moving from class weights to focal loss collapses the three-seed standard deviation from 0.0700 to 0.00007 — roughly three orders of magnitude, from “a coin-flip’s width of uncertainty” to “machine-precision-adjacent.” This is sharpened by comparing against the single-seed ablation study, which reported a BankSim baseline PR-AUC of 0.8330 (`ablation.json`, seed 42 only, class weights). That single number sits roughly two-thirds of one standard deviation above the three-seed class-weight mean of 0.7870 reported here — a concrete illustration of why the project’s own methodology treats single-seed differences smaller than ~ 0.01 PR-AUC as noise: an unlucky or lucky seed choice alone can move the class-weighted BankSim baseline by several hundredths. Under focal loss the same arm’s three-seed spread (0.8616 ± 0.00007) makes this kind of seed sensitivity essentially disappear.

This variance-collapsing property is also what wins focal loss its place in the champion pipeline. Table X reports `final_champions.json`’s validation-selection step, which compares the full feature set (**A**+**B**+**C** combined, i.e. arm **D**) under class weights against the same feature set under focal loss and selects whichever wins on validation PR-AUC — never on test. Focal loss wins the selection on three of the four datasets: BankSim, SAP Würzburg, and synth_erp; on IBM AML the two mechanisms tie exactly at the saturated ceiling (1.0000 vs. 1.0000) and the pipeline’s tie-break falls to class weights. Note that this selection step differs from the controlled arm-D comparison in that it also runs a 20-configuration hyperparameter search per candidate (skipped

TABLE X: Champion validation selection (final_champions.json), full feature set (topology+TGN+ordinary). Margin = focal − class-weight on validation. †selection flagged reliable=false (SAP Würzburg’s HPO was skipped by the small-sample guard). †synth_erp is non-evidentiary.

Dataset	Val (cw)	Val (focal)	Winner (margin)	Test (cw)	Test (focal)
BankSim	0.9489±0.0009	0.9515±0.0007	focal (+0.0026)	0.9395±0.0006	0.9364±0.0015
IBM AML	1.0000±0.0000	1.0000±0.0000	cw (tie, 0.0000)	1.0000±0.0000	1.0000±0.0000
SAP Würzburg	0.2888±0.0436	0.3562±0.0195	focal† (+0.0674)	0.1401±0.0076	0.1557±0.0717
synth_erp†	0.8487±0.0017	0.8507±0.0025	focal (+0.0020)	0.7722±0.0014	0.7785±0.0053

TABLE XI: TGN embeddings vs. hand-crafted topology, 3-seed test PR-AUC deltas (class-weighted arms only) and TGN’s share of total SHAP importance in the champion model. †synth_erp is non-evidentiary.

Dataset	TGN—base.	TGN—topo.	Comb.—topo.	TGN SHAP share
BankSim	+0.1326	−0.0123	+0.0004	17.5%
IBM AML	−0.0048	−0.0128	0.0000	1.9%
SAP Würzburg	−0.0987	−0.0124	+0.0684	42.0%
synth_erp†	+0.4489	+0.2404	+0.2850	37.3%

only for SAP Würzburg, whose validation fraud count trips the pipeline’s own small-sample guard), so its validation numbers are close to, but not identical with, the fixed-hyperparameter arm-D numbers reported elsewhere in this section — for example BankSim’s tuned class-weight validation PR-AUC is 0.9489 versus the fixed-hyperparameter arm D’s 0.9426.

BankSim deserves a specific flag: its validation margin, +0.0026 PR-AUC, is the thinnest *reliable, non-saturated, real-data* margin in the entire selection table — synth_erp’s nominal margin (+0.0020) is numerically smaller but is excluded from thesis claims as synthetic data, and IBM AML’s margin is a literal zero-width tie on an already-saturated metric, not a genuine comparison. A margin this thin, on the dataset the project treats as its cleanest structural-signal testbed, does not survive contact with the test set: the class-weight arm actually scores higher on BankSim’s held-out test period (0.9395 ± 0.0006) than the focal arm that was selected on validation (0.9364 ± 0.0015), a reversal of 0.0032 PR-AUC. We report this plainly as a caution against over-reading small validation margins, per the project’s own threshold-freezing discipline; it does not overturn the selection (validation-only selection is the whole point of never touching test until after the choice is frozen), and it is operationally invisible at the metric an auditor actually cares about — precision@100 is 1.00 for both arms.

2) *Temporal graph network embeddings*: Table XI isolates the value of the learned TGN representation against two references: the ordinary-only baseline (arm C — arm A, both class-weighted) and the hand-crafted topology block (arm C — arm B), plus the incremental value of combining both representations (arm D — arm B) and TGN’s share of total SHAP feature importance in the champion model (from final_champions.json’s per-dataset SHAP summary).

The central, load-bearing finding of this table is the middle column. On every real dataset, TGN alone (arm C) scores *below* hand-crafted topology alone (arm B): −0.0123 on BankSim, −0.0128 on IBM AML, −0.0124 on SAP Würzburg. These three deltas are remarkably tight given how

different the three datasets are in scale, fraud rate, and graph density — all three cluster within 0.0005 of −0.0125. Learned, self-supervised structural embeddings simply do not beat five purpose-built, domain-informed detectors on real transaction graphs in this study; the only dataset where TGN outperforms hand-crafted topology is synth_erp (+0.2404), the project’s own synthetic environment, which is explicitly excluded from evidentiary claims (Section IV-H). The first column tells a related but distinct story: TGN’s lift over the plain ordinary-feature baseline is positive on BankSim (+0.1326, nearly matching topology’s own +0.1449 lift there) but slightly *negative* on both IBM AML (−0.0048) and SAP Würzburg (−0.0987). On IBM AML this is a small movement within noise (both arms sit near the saturated ceiling, std ≈ 0.01); on SAP Würzburg it is a large, consistent penalty, echoing the same dataset’s negative topology lift reported in the ablation study — on this small, thin ERP ledger, adding structural feature complexity of either kind tends to hurt rather than help, plausibly because 25 test frauds cannot support the additional degrees of freedom either representation introduces.

Combining topology and TGN (arm D vs. arm B) adds negligible value beyond topology alone on the two datasets with the cleanest signal: +0.0004 on BankSim, 0.0000 on IBM AML (already saturated). SAP Würzburg’s combined arm does show a nominal +0.0684 recovery over its topology-only arm, but this sits inside the same small-sample volatility that produced SAP’s other large, sign-flipping deltas in this section and should not be read as evidence that TGN rescues topology’s failure there. synth_erp’s large combined gain (+0.2850) is, again, a synthetic-data result only.

This pattern is consistent with, not contradicted by, the 2026 literature on graph-native alternatives: Lei et al.’s STC-MixHop [29] reports the same qualitative result on a different transaction-fraud benchmark, that a graph-learning architecture struggles to outperform a tabular baseline once node attributes already carry strong signal, and attributes graph structure’s value specifically to settings where relational patterns are operationally significant rather than universal. This paper’s own dataset-by-dataset TGN comparison is, in effect, an independent empirical instance of exactly that claim: TGN loses to hand-crafted topology on every real dataset in this study, each of which differs in scale, fraud rate, and graph density, and wins only on the one dataset (synth_erp) explicitly engineered to have rich, easily learnable structure.

A last observation cautions against reading SHAP importance share as a proxy for ablation-validated value. SAP Würzburg’s TGN block claims the *largest* share of total SHAP importance of any dataset in this study (42.0%) despite producing the study’s most negative TGN lift (−0.0987 over baseline, −0.0124 over topology); IBM AML shows the opposite pattern, with TGN carrying almost no importance (1.9%) on a baseline it also fails to improve. A feature family can dominate a model’s internal attribution and still not earn its place in a leakage-free ablation — the same lesson the ablation study draws for hand-crafted cycle features applies here to a learned representation as well, via TreeSHAP’s exact per-feature attribution [37], [38].

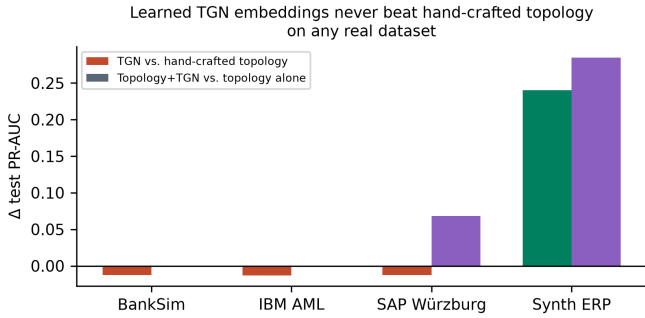


Fig. 9: TGN-vs-topology and combined-vs-topology PR-AUC deltas by dataset (modern.json). The TGN-vs-topology bar is negative on all three real datasets and positive only on the non-evidentiary synth_erp dataset.

3) *Implication for the production system:* Two independent findings close off TGN as a candidate for the served model, both already on the record from this build. First, TGN training is not bit-reproducible across machines even with a fixed seed: re-running the TGN trainer on freshly downloaded copies of the same source data and re-scoring against the committed champion bundle does not reproduce the committed test PR-AUC — BankSim recomputes to 0.845 against the bundle’s recorded 0.938, and SAP Würzburg to 0.098 against the bundle’s 0.140 (Section IV-H, Threats to Validity). Small floating-point differences in CPU-side PyTorch training compound across epochs and propagate through a feature block that carries 17–42% of the champion model’s total SHAP importance (Table XI), so an offline champion bundle trained on one machine cannot be assumed to reproduce its recorded score on another. Second, and independently disqualifying, the TGN’s per-account GRU memory tensor — the running state that makes its embeddings meaningful at any given point in time — is never persisted by the training code; it exists only inside the training process and is discarded on exit, so there is no artifact from which a live scoring service could resume the memory a freshly arriving transaction would need.

For these reasons, the production system described in Section III-H does not serve the TGN-augmented champion at all. It serves a separate, explicitly named *online* bundle — ordinary features plus hand-crafted topology only, arm B’s feature set — built and reported independently in `paper/data/online_bundles.json`: 34 features and test PR-AUC 0.9311 ± 0.0005 on BankSim, 33 features and 0.1094 ± 0.0198 on SAP Würzburg, and 28 features and 0.5702 ± 0.0174 on synth_erp. IBM AML’s online-bundle entry, labeled `ibm_aml_kaggle` (26 features, test PR-AUC 0.5120 ± 0.0674), is a fresh Kaggle re-download reproduction check, not the same simulator instance behind this section’s headline IBM AML numbers, and its PR-AUC is correspondingly far below this section’s saturated 1.0000 — a difference of instance, not of method, and one more reason the two IBM AML number sets in this paper must never be merged. In short: every PR-AUC reported in this section for the “champion” or “combined” arm describes an

TABLE XII: Realism audit: synth_erp vs. the real-dataset reference band (min-max across `ibm_aml`, `banksim`, `sap_wurzburg`, `paysim`). Source: `calibration_reference.json`, `synth_erp_realism.json`.

Metric	synth_erp	Real min	Real max	In band?
Fraud rate	0.4884%	0.1236%	1.2108%	Yes
Repeat-pair txn. share	0.8586	0.0	0.9936	Yes
Namespace overlap	0.9410	0.0	0.9927	Yes
loglp(amount) std.	2.3256	0.9551	2.0193	No (above)
Benford MAD	0.0047	0.0142	0.0758	No (below)
Pair reciprocity	0.1356	0.0	0.0446	No (3.0× above)

offline, research-only model; the number that describes what a live transaction would actually be scored against is the online-bundle figure above, and it excludes TGN entirely.

D. Synthetic Economy: Realism and Augmentation Results

For every statistic we report on synth_erp, we ask one question: does it fall inside the range spanned by the four real datasets (`ibm_aml`, `banksim`, `sap_wurzburg`, `paysim`), measured with identical methodology, or outside it? Table XII answers this for six statistics drawn from `calibration_reference.json` and `synth_erp_realism.json`. The fraud rate, the repeat-pair share of transactions, and the sender/receiver namespace overlap all land inside the real span. Three do not, and we report them plainly rather than tuning them away. Log-amount dispersion (`loglp_std` = 2.3256) sits above every real dataset (span 0.9551–2.0193): the simulated economy mixes cents-scale retail tickets with treasury sweeps of hundreds of thousands of dollars in one distribution, a wider mixture than any single-domain real dataset exhibits — genuine *over*-dispersion, not a defensible match. Benford mean-absolute-deviation (`benford_mad` = 0.0047) is *lower* than every real dataset (span 0.0142–0.0758): synth_erp’s amounts obey Benford’s law more cleanly than any real ledger we measured, itself a realism failure in the opposite direction — real accounting data carries idiosyncratic roundness and human-entered biases that a mixture of lognormal generators does not reproduce. Pair reciprocity (`pair_reciprocity` = 0.1356) is roughly 3.0× the real maximum (0.0446, `ibm_aml`): salary-to-purchase circulation and refund/cover behavior generate more back-and-forth pairs than any of the four real economies structurally permit. None of these three deviations is hidden in downstream use: they are why synth_erp is never treated as evidence for this paper’s central topology claim, only as training material and a stress test, per the usage policy at the top of this section.

A more direct realism concern is whether fraud amounts alone give fraud away — the specific artifact this project identified in `ibm_aml` elsewhere in this paper (fraud amounts distinctively *small*, amount-ratio 0.068, globally separable at ROC-AUC ≈ 0.91). On synth_erp, the median fraud amount (\$1,692.59) is 23.79× the median benign amount (\$71.14; the equivalent ratio in `banksim`, from `calibration_reference.json`, is 11.9×), and a univariate log-amount classifier reaches ROC-AUC 0.8078

overall (KS statistic 0.604 between the fraud and benign log-amount distributions) — at first glance a fraud population skewed to B2B-sized amounts in a retail-heavy economy. Restricted to the amount band where 93% of fraud actually lives (\$300–\$20,000, $n = 5,228$ fraud rows in band), that same classifier collapses to ROC-AUC 0.4911 — indistinguishable from chance. Fraud amounts are globally distinguishable only because fraud skews toward a coarser amount regime than the bulk of retail traffic; *within* its own regime, amount carries essentially no signal, forcing any detector toward behavioral or structural evidence rather than a one-column amount rule — the opposite of the `ibm_aml` artifact, and a deliberate design target corroborated here after the fact rather than asserted in advance.

1) *Adversarial Tabular-Leak Probe*: A realistic amount distribution does not rule out a cheap categorical shortcut elsewhere in the table — the same class of hidden bias that motivates auditing tabular fraud-evaluation datasets for shortcut features before trusting them [55]. We therefore ran an automated probe over 16 candidate rules — every `tx_type` indicator, an account-creation-date indicator on each side of the transaction, and pairwise conjunctions of the two — scoring each rule’s support, precision, recall of fraud, and lift over the 0.4884% base rate. Table XIII reports the eight highest-lift rules out of the 16 tested. Three flow types (`tx_type=payroll`, `tx_type=recurring_bill`, `tx_type=refund`) and their conjunctions with a mid-year account carry *zero* fraud whatsoever (lift 0.0) — no scenario in the generator routes any of the five typologies onto these three rails at all, which is a genuine (if narrow) gap in coverage rather than a realism strength.

The rule the probe itself records as worst, `tx_type=transfer & dest_created_midear`, reaches a lift of $19.0\times$ the base rate, support of 26,420 transactions, precision of 9.29%, and recovers 43.75% of all fraud in a single two-clause rule. The dataset’s own automated gate does not classify this as a leak (`leak_flag = false`, its precision bar sitting well above what this rule reaches), and a rule flagging under 10% of the transactions it fires on as fraud is far from the near-deterministic, one-line shortcuts the generator’s earlier drafts contained before an adversarial review removed them. Still, a rule with $19\times$ lift and 44% recall is a real, exploitable regularity, not a non-finding, and we report it rather than let a binary pass/fail flag stand in for a reader’s own judgment of where “legitimate risk factor” ends and “tabular leak” begins. A transaction moving through the `transfer` rail toward an account opened within the same year is a defensible real-world risk heuristic — mid-year account creation correlates with fraud in practice, not only in this simulation — but we flag it here precisely so that any model trained on `synth_erp` can be checked for whether it quietly learned this shortcut instead of the intended structural and behavioral signal.

2) *The Augmentation Transfer Probe*: The realism and leak audits above establish what `synth_erp` is calibrated well enough to be used *for*. The most direct test of the narrower, practical question — does adding `synth_erp` rows to a

TABLE XIII: Adversarial tabular-leak probe, top 8 of 16 rules by lift. Base fraud rate 0.4884%. Source: `synth_erp_realism.json`, `tabular_leak_probe`.

Rule	Support	Precision	Recall	Lift
<code>transfer & dest_midear[†]</code>	26,420	0.0929	0.4375	19.0
<code>dest_created_midear</code>	67,621	0.0502	0.6049	10.3
<code>tx_type=transfer</code>	94,781	0.0385	0.6500	7.9
<code>vendor_payment & dest_midear[†]</code>	25,781	0.0316	0.1451	6.5
<code>sale & dest_midear[†]</code>	8,131	0.0146	0.0212	3.0
<code>src_created_midear</code>	156,550	0.0113	0.3147	2.3
<code>reimbursement & dest_midear[†]</code>	741	0.0081	0.0011	1.7
<code>tx_type=reimbursement</code>	12,236	0.0079	0.0173	1.6

[†]`dest_midear` abbreviates `dest_created_midear`.

TABLE XIV: Augmentation transfer probe: XGBoost PR-AUC, 3-seed mean $\pm$ std. `synth_only` is zero-shot (trained solely on `synth_erp`, evaluated on the real test split). Source: `augmentation.json`.

Dataset	Real-only	Real+synth	Δ	Synth-only
<code>banksim</code> ($n=1320$)	0.8669 ± 0.0031	0.7610 ± 0.0022	-0.1059 ± 0.0011	0.7243 ± 0.0079
<code>ibm_aml</code> ($n=377$)	0.99999 ± 0.00001	0.9954 ± 0.0013	-0.0046 ± 0.0013	0.0017 ± 0.0006
<code>sap_wurzburg</code> ($n=25$)	0.1366 ± 0.0446	0.0293 ± 0.0033	-0.1073 ± 0.0417	0.0008 ± 0.0001

real dataset’s training split make its fraud model better? — is a paired augmentation experiment: for each of three real datasets, train an XGBoost model [60] three ways (real data only; real data with all 1,148,503 `synth_erp` rows pooled into the training split; and `synth_erp` data only, evaluated zero-shot on the real test split), each averaged over three seeds, under the same time-aware split, class-weight imbalance handling, and validation-tuned threshold used throughout this paper. Table XIV reports the result, and it is unambiguous: pooling synthetic rows into training *hurts* test PR-AUC on every one of the three real datasets tested. `banksim` falls from 0.8669 ± 0.0031 (real-only) to 0.7610 ± 0.0022 (real+synth), a paired per-seed delta of -0.1059 ± 0.0011 ; `ibm_aml` falls from an already-saturated 0.99999 ± 0.00001 to 0.9954 ± 0.0013 , $\Delta = -0.0046 \pm 0.0013$ — small in absolute terms, but the baseline leaves almost no headroom to fall into, so even this small drop is a consistent, same-direction result; and `sap_wurzburg` — on only 25 test-set frauds, the same small-sample caveat that applies to every `sap_wurzburg` result in this paper — falls from 0.1366 ± 0.0446 to 0.0293 ± 0.0033 , $\Delta = -0.1073 \pm 0.0417$. The mechanism is unsurprising in hindsight given Sec. IV-D: a single tree ensemble fit jointly over two economies with measurably different amount dispersion and reciprocity structure pulls its splits toward the pooled distribution, at the real dataset’s expense. We report this as a clean negative result rather than qualify it away: *naive* pooled augmentation with this dataset, under this protocol, does not help, on any real dataset we tested it against.

There is exactly one genuinely positive number in this experiment, and it is a narrower, different claim than “synthetic data helps supervised training”: the `synth_only` zero-shot column for `banksim`. A model that has *never seen a single real BankSim row* — trained entirely on `synth_erp`, evaluated as-is on real, held-out `banksim` test data — scores 0.7243 ± 0.0079 PR-AUC, recovering

83.6% of the 0.8669 real-trained baseline, at precision@100 of 0.99 ± 0.0082 against the real-trained baseline’s perfect 1.00; chance at this split’s 1.21% fraud rate is on the order of 0.01–0.02 PR-AUC, so this is far from a vacuous transfer. The same zero-shot comparison fails on `ibm_aml` (0.0017 PR-AUC against a 0.99999 real-trained ceiling — unsurprising, since `ibm_aml`’s giveaway-small fraud amounts are inverted relative to `synth_erp`’s B2B-skewed fraud) and on `sap_wuerzburg` (0.0008 PR-AUC, 25 test frauds, too small to support any transfer claim). We read the `banksim` result as evidence that the synthetic economy teaches *transferable structural and behavioral discrimination* that generalizes to a real domain it structurally resembles — not as evidence that pooling synthetic rows into a real training set is a useful augmentation recipe. These are two different claims, only one of which this experiment supports; we are careful not to let the positive framing of the zero-shot result launder the negative framing of the pooled-augmentation result three rows above it.

3) *Summary and Scope*: `synth_erp` is a large, ERP-flavored, multi-typology synthetic ledger built under a binding non-circularity constraint and audited, after the fact, against four real datasets on realism, tabular leakage, and downstream transfer. It matches the real fraud-rate, repeat-pair, and namespace-overlap bands; it is measurably over-dispersed in amount and reciprocity and unrealistically Benford-clean relative to every real dataset compared against it; its cheapest categorical shortcut reaches $19\times$ lift and 44% recall without being flagged as a formal leak, a judgment call we surface rather than hide; and pooling it into real training data hurts every real dataset’s PR-AUC we tested, its one positive contribution being a narrower zero-shot transfer result on the one real dataset it structurally resembles. Consistent with the usage policy stated at the outset, none of this section’s numbers are used anywhere in this paper as evidence for the central topology claim (Sec. IV-B); `synth_erp`’s role is training material, a per-typology stress test, and demo content for the audit console (Sec. III-H), never proof.

E. Explainable Audit Alerts: Results

Three of the four datasets reach perfect precision at their operating k (Table IV): every one of BankSim’s top 15, IBM AML’s top 20, and `synth_erp`’s top 20 test-period alerts is a labelled fraud. IBM AML’s 1.00 should not be over-read, however: its baseline model is already saturated at PR-AUC ≈ 0.99 (the ablation study reported earlier in this paper), so a perfect top- k here is consistent with the amount signal alone doing the work and does not, on its own, demonstrate that topology mattered for these particular alerts — three of the twenty carry no reconstructable fan-in or cycle evidence at all (`evidence_type_counts.none=3`), meaning their risk drivers were purely ordinary/tgn.

SAP Würzburg is the outlier the ablation predicts: precision@10 is 0.20 (2 of 10), the weakest by a wide margin, the direct operational consequence of the dataset’s negative marginal structural lift reported elsewhere in this paper, compounded by only 25 labelled test frauds. Every one of its

TABLE XV: Top-3 globally attributed features per dataset by mean |SHAP| (margin space), with family tag (`paper/data/final_champions.json`).

Dataset	Rank	Feature (family)
BankSim	1	category_es_transportation (ordinary)
	2	fan_in (topology)
	3	dest_in_degree (topology)
IBM AML	1	lap_out_recur (topo.-amt-derived)
	2	log_amount (ordinary)
	3	lap_out_payees (topo.-amt-derived)
SAP Würzburg	1	fan_in (topology)
	2	dest_count_prior (ordinary)
	3	log_amount (ordinary)
synth_erp	1	log_amount (ordinary)
	2	tgn_edge_expectedness (tgn)
	3	dest_in_degree (topology)

ten alerts is nonetheless a syntactically well-formed, replay-verified fan-in – destinations collecting from 166 to 271 distinct senders in-window – because those accounts are simply high-traffic general-ledger hubs (GL: 895000, GL: 300000), not because fraud is present. This is the clearest illustration in the paper of what Layer 2 actually guarantees: the evidence is provably consistent with what the model saw, never a guarantee that what the model saw was fraud. (Its test-partition size also differs by source file: `final_champions.json`, which produced this bundle, reports 40,116 test rows / 25 frauds, while `ablation.json`’s separately-run partition-aware split reports 40,139 – the two pipeline runs are not the same split and should not be merged.)

1) *Global Attribution Is Dataset-Specific, Not Universal*: Averaging |SHAP| over a sample of each dataset’s test rows (`final_champions.json`: `shap_global_top20`, 50,000 sampled rows for BankSim/IBM AML/`synth_erp` and the full 40,114-row test partition for SAP Würzburg, since `bundle_metadata.json` does not itself carry per-feature magnitudes) shows that no single family dominates across datasets (Table XV).

BankSim’s single most-attributed feature globally is a merchant-category indicator, not a topology feature – `fan_in` and `dest_in_degree` rank second and third – even though BankSim is the dataset with the strongest validated structural lift in this paper. Amount and category signal dominates the average attribution even where topology genuinely helps; the two findings are compatible because global mean |SHAP| and marginal ablation lift answer different questions (which features move scores the most, vs. how much test-set PR-AUC a feature group adds).

IBM AML’s global top feature is `lap_out_recur`, tagged topology-amount-derived rather than pure topology – confirming at the dataset level the alert-level pattern noted above, and consistent with `final_champions.json`’s own recorded caveat for this dataset (“baseline is amount-saturated (PR-AUC ~ 1.0): margins here are uninformative”). No genuine cycle feature appears in its top 20 at all, despite cycles accounting for

8 of its 20 alert-evidence blocks – the model leans on the amount-derived lapping proxy while Layer 2’s independent replay still finds and shows a real closed loop.

SAP Würzburg is the only dataset whose global top feature is a *pure* topology column (`fan_in`), yet it also has the negative marginal structural lift and the worst alert precision in this paper: mean $|\text{SHAP}|$ measures how much a feature moves the score on average, not whether that movement is correct, and `fan_in` can be the model’s largest lever while remaining a poor discriminator when the underlying counts reflect a handful of high-traffic GL hub accounts rather than fraud. `synth_erp` – sanity/demonstration data, never thesis evidence – shows a third pattern: `log_amount` leads, with a learned TGN feature and a topology feature close behind, echoing the earlier finding that on this dataset learned and hand-engineered structure are complementary rather than either dominating.

Together, Tables IV and XV argue against any single sentence like “the model relies on graph structure”: which family dominates, and whether that reliance is well-founded, is dataset-specific. The console’s honesty about that – visible family tags, replay-verifiable evidence, and a `ground_truth_label` kept for research transparency but never surfaced to a real reviewer – is the point of the two-layer design, not a substitute for the ablation evidence that grounds it.

F. The Online/Offline Serving Gap

The champion models evaluated earlier in this paper (Section V) include a Temporal Graph Network (TGN) embedding stage [24] feeding an XGBoost head [60]. These champions are *not* servable online, and the paper makes no claim that they are. Two independent problems rule it out. First, TGN inference requires a memory tensor built by replaying a transaction sequence through the trained network; nothing in this build persists that tensor, so scoring a single live transaction would require replaying the dataset’s entire prior history on every request — the opposite of the online engine’s $O(1)$ -per-feature design. Second, and more fundamentally, TGN training on this codebase is not bit-reproducible across machines even with a fixed seed: re-running `modeling.tgn` on freshly downloaded, same-source data and re-scoring against the committed champion bundle diverges substantially (documented in `docs/BUILD_SPEC.md` as a reproduction failure on both `banksim` and `sap_wurzburg` test folds), because small floating-point differences in CPU training compound over epochs through a component that accounts for 17–42% of the champion’s feature importance. A component that will not reproduce its own training run on the same machine is not a component a production system should be handed authority to persist and reload.

The deployed artifact is therefore a deliberately smaller “online” bundle, trained on ordinary and topology features only, with the TGN family dropped entirely rather than approximated. This is verifiable directly in the two bundle manifests: the online bundle’s feature count is, dataset by dataset, exactly the ordinary-plus-topology subtotal of the champion’s own feature-family breakdown (`banksim`: $21 + 13 = 34$;

TABLE XVI: Test PR-AUC: offline TGN-inclusive champion vs. the deployed online (ordinary+topology only) bundle, same dataset instance, 3 seeds each.

Dataset	Champion	Online	Gap
<code>banksim</code>	0.93637 ± 0.00154	0.93114 ± 0.00045	0.00523
<code>sap_wurzburg</code>	0.1557 ± 0.07166	0.10939 ± 0.01982	0.04631
<code>synth_erp</code>	0.77845 ± 0.00526	0.57018 ± 0.01744	0.20827

`sap_wurzburg`: $20 + 13 = 33$; `synth_erp`: $15 + 13 = 28$; against champion totals of 99, 98, and 93 respectively, each with 65 TGN dimensions removed) — confirming the online bundle is the champion’s own feature set with TGN excised, not an independently retrained approximation. Table XVI reports what that exclusion costs in test PR-AUC, computed directly from `paper/data/final_champions.json` (champion) and `paper/data/online_bundles.json` (online).

The cost of going online is small on `banksim` (0.005 PR-AUC) but is not uniformly small: `synth_erp` pays roughly 0.21 PR-AUC and `sap_wurzburg` roughly 0.05, consistent with a component contributing 17–42% of feature importance being removed entirely. `sap_wurzburg`’s champion figure also carries the small-test-fold caveat attached to that dataset elsewhere in this paper — both its ablation split and its champion split contain only 25 fraud cases in the test fold, so both the champion and online numbers for that dataset should be read as high-variance point estimates, not precise costs. IBM AML is deliberately omitted from Table XVI: the champion figure in `final_champions.json` (`ibm_aml`, PR-AUC 1.0 ± 0.0 , consistent with the baseline-saturation finding reported elsewhere in this paper) is trained on a different underlying data acquisition than `ibm_aml_kaggle` (PR-AUC 0.512 ± 0.06743), the fresh Kaggle re-download the online bundle is actually trained and served on. Reporting a “gap” between two different simulator instances would misstate the cost of removing TGN as the cost of a different dataset entirely, so no such number is reported; `ibm_aml_kaggle`’s figure stands on its own as the deployed bundle’s performance on the data it was actually built from.

A second, smaller and more deliberate divergence exists between the two paths’ ordinary-feature computation. The batch engine’s expanding per-account z-score uses the textbook two-pass variance formula $(\bar{x}^2 - \bar{x}^2)$, which suffers catastrophic cancellation for accounts whose historical amounts are nearly identical — verified on the full `banksim` dataset to produce spurious $|z| > 1000$ values (observed up to $\sim 10^8$) on 1,238 of 594,643 rows (0.21%), where a numerically stable computation should return a near-zero or undefined value. `app/services/online_ordinary.py` computes the same statistic via Welford’s algorithm, which does not exhibit this failure mode and instead returns NaN. Rather than reproduce a known instability from the locked research pipeline into new serving code for the sake of train/serve identity, the online path uses the numerically correct formulation and accepts a small, disclosed, and precisely quantified skew on 0.21% of rows as the honest cost of not shipping a bug on purpose.

Query	Before	After	Speedup
Paid invoices, newest-first (status='paid' ORDER BY invoice_date DESC LIMIT 50)	Bitmap Heap Scan → top-N heaport: 875 buffer hit: 6.331 ms	Index Scan Backward, no sort; 50 buffer hit: 0.301 ms	~21x
Alert queue (ORDER BY score DESC LIMIT 50)	No index existed; seq-scan-and-sort base- line never captured at scale?	Index Scan (DESC); 0.236 ms	—
Vendor-scoped invoices (vendor_id=7 ORDER BY id LIMIT 50)	Bitmap Index Scan on vendor_id + small sort: 0.253 ms	Same plan — new composite index not selected: 0.317 ms	none

TABLE XVII: Module 7 query optimization, before/after EXPLAIN (ANALYZE, BUFFERS) on a bulk-seeded dev database. Row 3 is reported as a genuine non-win, not silently dropped.

G. Measured Performance, Concurrency, and Recovery

Query optimization was driven by captured EXPLAIN (ANALYZE, BUFFERS) plans against a reproducibly bulk-seeded dev database (5,000 vendors, 40,000 invoices, 23,929 payments, 5,000 audit alerts; seed 42; ANALYZE re-run after load so planner statistics are current), not guessed indexes. Table XVII reports the result honestly, including the one index that did not help: a composite index added to the vendor-scoped invoice listing on the expectation that it would repeat the first query’s win did not change the query plan at all, because at this dataset’s distribution (roughly 8 invoices per vendor) sorting that few rows is already cheaper than the planner switching indexes. It remains in the schema — cheap to maintain, and the correct choice the moment a single vendor’s invoice volume becomes skewed — but is reported as a non-win rather than silently dropped or claimed as a success it was not.

The audit_alerts table was empty during earlier module testing; no pre-index baseline exists at production row counts, so degradation was inferred, not measured.

Redis-backed response caching (route+params+tenant+role key, ETag/304 support, explicit invalidation on write) and all four new indexes ship as ordinary CREATE INDEX migrations rather than CONCURRENTLY builds — an acceptable simplification against an empty or lightly loaded dev-scale table, but a production rollout against a live, concurrently written table should use postgresql_concurrently=True outside a transaction block to avoid holding write locks for the duration of the index build.

Observability runs as a verified Prometheus/Grafana/Uptime Kuma stack, not a planned one. Prometheus scrapes /metrics every 10 seconds with the shadow-ledger-api target confirmed healthy; a Grafana dashboard (“Shadow Ledger — Overview”) is auto-provisioned alongside the datasource and exposes request rate and 5xx rate by route, p95 latency by route, fraud-scoring latency p50/p95/p99, cache hit rate, and duplicate-payment attempts. fraud_scoring_latency_seconds{dataset} is the histogram the paper’s sub-200 ms p95 latency target is measured against, reflecting the online path’s O(1)-per-feature design (Section IV-F): topology lookup, Welford-based ordinary features, one small XGBoost DMatrix predict, and TreeSHAP, with no whole-stream replay on the request path. Uptime Kuma probes /health and /ready every 30 seconds; /ready already returns 503 the instant the database, Redis, or the scoring registry is unavailable, so the monitor’s job is only to alert on a non-200, not to implement the health logic itself. Error logging is structured JSON with correlation-id propagation from the access log through every service call (app/core/logging.py), live

TABLE XVIII: Module 9 concurrency and recovery drills — measured results.

Drill	Configuration	Result
Duplicate-payment race	k6, 30 VUs, 1 invoice, distinct idempotency keys	1/30 succeeded, 29/29 rejected (duplicate_payment); DB confirms exactly 1 payment row; successful-request latency avg 267 ms, p95 807 ms
Mixed workload ramp	k6, 0→50→200→500 VUs, single uvicorn worker, 40-conn. DB pool	39,727 requests attempted, 28,642 completed; overall p95 8.18 s; error rate 4.42% (<5% threshold passed; p95 < 1 s threshold failed)
Backup/restore drill	pg_dump -Fc → clean throwaway DB; row-count + payment-total fingerprint compare	Quiescent run: match=true, RTO= 3.07 s (dump 0.95 s/6.99 MB, restore 2.12 s); a run taken concurrently with active writes legitimately reports match=false (Postgres MVCC), reproduced again in the most recent committed drill

since the project’s first module and the actual mechanism every bug found during query optimization and testing was diagnosed from; a self-hosted GlitchTip instance was scoped but not deployed, since it requires its own Postgres database, Redis, and Celery worker rather than a docker-compose add-on line — GLITCHTIP_DSN is already read by app/core/config.py as a drop-in point once a target instance exists.

Concurrency, load, and recovery were exercised as k6 scripts and a scripted backup/restore drill, with every figure below taken from an actual run (docs/TESTING_MODULE9.md), not a projection. The duplicate-payment race (tests/load/payment_race.js) is the automated form of the defense in Section III-H2: 30 virtual users fire truly concurrent payment requests at one invoice, each with a distinct idempotency key, so that the only thing standing between them and a double payment is the row lock and unique index, not idempotency replay. The mixed-workload ramp (tests/load/mixed_workload.js) drives a 70% browse / 20% invoice-submission / 10% alert-queue-read mix from 0 to 500 concurrent users against a single dev-mode uvicorn process with no worker pool and a 40-connection database pool, deliberately reported as a degradation-point measurement rather than a pass/fail gate.

The mixed-workload result is reported as found: the system clearly degrades before reaching 500 concurrent users, with I/O timeouts appearing during the highest-VU stage and overall p95 latency an order of magnitude above the 1-second threshold. This is expected of a single dev-mode process and is the useful output of the test — it names horizontal scaling (--workers N, a larger pool, or multiple instances behind a load balancer) as the concrete next step for a production rollout, rather than asserting a latency number the current deployment configuration cannot support. The backup/restore drill’s own history is a second methodology finding worth stating rather than hiding: an initial drill run taken while the mixed-

TABLE XIX: Threats to validity, at a glance. Every magnitude is read directly from a named artifact file, never re-derived here.

Threat	Magnitude / evidence	Disclosure
Single-seed ablation, no CIs	IBM AML full-model lift +0.00792 is below the project’s own ~ 0.01 noise floor	§IV-B2; 3-seed champion partially corroborates
TGN not bit-reproducible	BankSim 0.938 $\rightarrow$ 0.845; SAP W. 0.140 $\rightarrow$ 0.098, fixed seed, same source data, different machine	docs/BUILD_SPEC.md; §IV-F
Batch z-score instability	1,238/594,643 (0.21%) BankSim rows, $ z > 10^3 - 10^8$ from catastrophic cancellation	Online engine uses Welford; batch pipeline uncorrected
Synthetic augmentation hurts	Δ PR-AUC $-0.1059/-0.00455/-0.10725$ on 3 real datasets, all 3 seeds each	Never used in the deployed bundle’s training data
SAP W. small test fold	25 test frauds; 3-seed champion std ± 0.0717 on mean 0.1557 (46% rel.)	Flagged high-variance at every occurrence
Deployed $\neq$ champion	Online-bundle gap 0.005–0.208 PR-AUC by dataset	Table XVI; every PR-AUC labeled champion/online
Coarse RBAC, single tenant	3 roles, no maker-checker chain; no cross-org isolation tested	Named explicitly as un-evaluated scope

workload test was still writing to the database correctly reported a fingerprint mismatch, not because backup or restore is broken, but because `pg_dump` does not block concurrent writers under Postgres MVCC, so a fingerprint query issued moments before the dump can legitimately diverge from what the dump itself captures. The most recently committed drill artifact (`infra/backups/last_drill_report.json`) reproduces exactly this pattern (`match=false`, RTO 2.85s) against live traffic, while the clean, quiescent-database run reported in Table XVIII passed. The operational implication is direct: a production backup schedule needs either a quiescent window, a shared `pg_export_snapshot()`, or reliance on the continuous WAL archiving already enabled (`archive_mode=on`) for point-in-time recovery — none of which this session exercised end-to-end, and all of which are named as the concrete next drill rather than assumed solved.

H. Discussion: Threats to Validity and Limitations

The preceding sections report several results this paper is obligated, by its own stated methodology, never to soften: a negative topology lift on SAP Würzburg, an uninformative comparison on a saturated IBM AML baseline, a single-seed ablation, and a deployed model that is provably not the model evaluated as the research champion. This section collects those admissions in one place, cross-referenced rather than re-derived, and adds several threats not yet raised elsewhere in the paper: the measured cost of naive synthetic augmentation, a numerical-stability defect that quietly touches every headline number in this study rather than only the serving path, and the scope boundaries of the production system’s access-control and deployment model. Table XIX summarizes all seven; the remainder of this section explains each.

1) *Statistical power: one seed is not an estimate*: The seven-arm ablation that produces this paper’s headline lift numbers (Table VII) is a single training run per arm, seed 42, with no repeated trials, no confidence intervals, and no significance test of any kind. §IV-B2 already applies the project’s own ~ 0.01 PR-AUC noise-floor heuristic to this

fact and concludes that IBM AML’s full-model structure lift (+0.00792) cannot be distinguished from zero on the evidence available; we restate it here as what it is: not a p-value, but an informal threshold the project’s own harness code uses to decide when a difference is worth interpreting at all. No formal bootstrap or paired test has been run anywhere in this study to replace that heuristic. The separate three-seed champion runs in `final_champions.json` corroborate the qualitative picture — BankSim stable at 0.9364 ± 0.0015 , SAP Würzburg volatile at 0.1557 ± 0.0717 — but they evaluate a different, TGN-inclusive feature set on a (mostly) different split convention, so they are corroboration, not a repetition of the ablation itself. SAP Würzburg’s 25 test-split frauds are, on their own, close to a hard ceiling on what any single-dataset PR-AUC estimate on this data can support; we treat its negative lift as directionally credible but numerically imprecise throughout, never as a precise point estimate. Even the split denominators behind this result are not perfectly reconciled: `ablation.json` and `final_champions.json` report two slightly different partition-aware split sizes for SAP Würzburg (120,411/40,137/40,139 vs. 120,341/40,114/40,116, a 0.06% difference, §III-D), traced to separate ETL passes but not further diagnosed.

2) *Two disclosed non-reproducibilities*: Two independent findings surfaced during this build’s own reproduction attempts, both load-bearing enough to affect how every TGN-inclusive number in this paper should be read. First, re-training the TGN encoder on freshly downloaded copies of the same source data, with the same fixed seed, does not recover the committed champion bundle’s test PR-AUC: BankSim recomputes to 0.845 against the bundle’s recorded 0.938, and SAP Würzburg to 0.098 against 0.140 (`docs/BUILD_SPEC.md`; §IV-F). CPU-side floating-point differences in PyTorch training compound across epochs and propagate through a feature block carrying 1.9–42.0% of the champion model’s SHAP importance depending on dataset. Practically, the champion PR-AUC figures reported in this paper are this machine’s own measurement, not a constant guaranteed to reproduce on different hardware with the same seed and code — a property one would not expect to have to state for a gradient-boosted tree, but must, because the TGN stage feeding it is not deterministic across machines.

Second, and unrelated to cross-machine reproducibility, the batch feature pipeline’s expanding per-account z-score uses the textbook two-pass variance formula ($\bar{x}^2 - \bar{\bar{x}}^2$), subject to catastrophic cancellation for accounts whose historical amounts are nearly identical. Quantified on the full BankSim dataset, this produces spurious $|z| > 1,000$ values (observed up to $\sim 10^8$) on 1,238 of 594,643 rows (0.21%), where a stable computation returns a near-zero or undefined value. This is not merely a serving-time footnote: the identical formula runs across all four datasets’ batch builds, so every headline ablation and champion PR-AUC in this paper — not only the BankSim figure it is quantified on — was computed with this instability present somewhere upstream, at an unmeasured rate on the other three datasets. The online engine (`app/services/online_ordinary.py`)

uses Welford’s algorithm instead, which does not fail this way; this was a deliberate choice to ship correct code going forward rather than replicate a known bug for train/serve identity, and it is the one place the offline and online paths diverge by construction rather than by TGN’s absence alone. The batch pipeline itself remains uncorrected – fixing it would move this paper’s own headline numbers after the fact – so we disclose the defect precisely rather than patch it mid-study.

3) *Synthetic data: barred as evidence, and measurably harmful as augmentation:* synth_erp is this project’s own generated corpus, and the project’s binding policy treats it as a stress-testing environment only, never as thesis evidence – precisely to avoid grading its own homework with data built to contain the structure the paper argues for. Its largest-in-study topology lift ($+0.4031$ PR-AUC, 246.8% relative) is reported in §IV-B under that explicit caveat and is not treated as support for this paper’s central claim.

A second, distinct question – does adding synth_erp rows to a real dataset’s training set help a real model – was tested directly (paper/data/augmentation.json, three seeds, 1,148,503 synthetic rows appended to each real training split) and answers plainly: it does not. Augmentation reduces test PR-AUC on every one of the three real datasets, at every seed: BankSim -0.1059 ± 0.0011 ($0.8669 \rightarrow 0.7610$), IBM AML -0.00455 ± 0.0013 ($0.99999 \rightarrow 0.99544$), and SAP Würzburg -0.1073 ± 0.0417 ($0.1366 \rightarrow 0.0293$). This is the more actionable of the two synthetic-data warnings: not only is synth_erp barred as evidence, but naively mixing it into a real model’s training data measurably degrades that model, plausibly by shifting the decision boundary toward synthetic-specific regularities that do not hold on real transactions – which is why the deployed online bundle (§IV-F) is trained exclusively on real data. One number in the same experiment cuts the other way and answers a narrower question: a model trained *only* on synth_erp, with zero real BankSim rows, still reaches PR-AUC 0.7243 ± 0.0079 scored zero-shot against real BankSim test data (versus 0.8669 ± 0.0031 real-trained) – evidence the generator’s structural realism transfers to some degree, a claim about generator fidelity rather than about whether *mixing* synthetic rows helps a real deployment. We report both rather than only the one that reads more favorably.

4) *The deployed model is not the evaluated champion:* Every PR-AUC attributed to a “champion” in this paper describes an offline, TGN-inclusive model that is not, and by construction cannot be, served online (§IV-F): TGN’s per-account memory tensor is never persisted, and TGN training is the non-reproducible component above, so a production system should not be handed authority to reload it. Table XVI quantifies what removing TGN costs against the same dataset instance: a small 0.00523 PR-AUC gap on BankSim, a larger 0.04631 on SAP Würzburg, and a substantial 0.20827 on synth_erp – proportional to how much of that dataset’s champion SHAP importance TGN carried (17.5%, 42.0%, 37.3%). IBM AML is excluded from that comparison because its online bundle trained on a separate, fresh Kaggle re-download (ibm_aml_kaggle), not the simulator instance behind its champion figure, so no single gap number honestly describes it. The risk this creates is real: the abstract’s headline

lifts (+11.7% BankSim, +246.8% synth_erp, non-evidentiary) both describe the offline champion, and a reader who quotes them as the deployed console’s accuracy would overstate it on every dataset except BankSim, where the gap is small.

5) *System scope: access control and deployment breadth:* The audit console enforces three named roles – `erp_clerk`, `forensic_auditor`, `system_administrator` – through one route-level guard (`app/core/deps.py:require_role`), surfaced through two front-end experiences: clerk-facing ERP screens that post invoices, vendors, and payments, and an auditor-facing console that reviews alerts. This is coarse by design: no maker-checker approval chain (a second clerk or auditor co-signing a payment above a threshold), no per-vendor or per-amount delegation, and no attribute-based policy beyond the three fixed roles. The deployment is single-tenant: the scoring registry keys one `FraudScoringService` instance per dataset, standing in for a client or business unit in this build, but no cross-tenant data isolation, per-tenant model versioning, or multi-organization quota behavior was implemented or tested. Federated training across institutions (a FedProx-style scheme letting organizations improve a shared model without exchanging raw transaction data) was scoped only as a future direction in the source project and was never implemented, benchmarked, or evaluated here; nothing in this architecture should be read as evidence toward a federated claim. A 2026 industry-scale study of exactly this deployment pattern, Roth et al.’s evaluation of federated fraud detection with NVIDIA FLARE [46], reports federated training closing most of the gap to a fully centralized model (F1 0.903 federated versus 0.925 centralized, against 0.643 for independently trained per-institution models) while never sharing raw transaction data across institutions – concrete external evidence that this paper’s own single-tenant scope is a real, addressable limitation rather than an architectural dead end, should this system ever need to serve more than one organization’s ledger.

a) *Regulatory and deployment context:* This paper’s explainability and audit-trail design choices are not made in a vacuum: Faccia’s 2026 study of explainable AI in auditing [41] and Uddin and Aziz’s 2026 evaluation of explanation reliability under U.S. regulatory compliance requirements [42] both argue, independently of this paper, that a fraud-detection system’s deployability depends on exact, stable, tree-native attribution at least as much as on raw PR-AUC – the same conclusion this paper’s own choice of TreeSHAP over an end-to-end neural architecture is built around (§III-D). Read against that literature, the console’s evidence-reconstruction layer (§IV-E) and its family-tagged attribution are best understood not as a cosmetic addition to a scoring model, but as the specific property a regulator-facing deployment of this system would need to demonstrate first. Two narrower validity gaps round out the picture: the four modeled datasets are static historical snapshots – none model an adversary adapting after observing what the deployed model flags, a concern the tabular-evaluation literature increasingly treats as first-class rather than an afterthought [55] – and every arm in this study is a gradient-boosted tree with at most one learned-graph

baseline (TGN), considerably narrower than the architecture space the broader graph-fraud literature explores [1], [2].

V. CONCLUSION

A. The claim this paper actually supports

This paper set out to answer one question honestly, in either direction: does leakage-free graph topology add predictive value over a strong transaction-only baseline? The answer, on the evidence gathered across four datasets under one shared, leakage-audited protocol, is conditional rather than universal, and we state it exactly as measured rather than rounded toward a cleaner story. On BankSim, topology helps decisively: a +0.0974 PR-AUC full-model structure lift (11.7% relative over a 0.833 baseline), independently confirmed by a clean, amount-blind probe that still finds +0.1899 PR-AUC once every amount feature is stripped from both sides of the comparison. On the project’s own synth_erp corpus, topology helps by the largest margin measured in this study (+0.4031, 246.8% relative) – but this number is explicitly barred from standing as thesis evidence by the project’s own policy, since the corpus was built to contain the structure being tested for, and stands only as a demonstration of what topology *can* do under favorable conditions. On IBM AML, the comparison that matters is uninformative rather than negative: an already-saturated 0.990 baseline leaves almost no headroom, the nominal full-model lift (+0.00792) sits below this project’s own noise floor, and the only honest positive signal is a narrower, amount-blind probe (+0.0905) that isolates a real but modest structural contribution once the dominant amount signal is removed from both arms. On SAP Würzburg, topology does not help – -0.08288 PR-AUC, a genuinely negative result on this project’s real general-ledger export, reported plainly rather than explained away, with the caveat that only 25 test-split frauds make this specific number imprecise even as its direction is corroborated by an independent, differently-built experiment: a 46% relative standard deviation on the 3-seed champion PR-AUC for the same dataset (§IV-H1). Four datasets, two clear directions and two qualified ones: this is the paper’s actual result, and it is deliberately not tidier than that.

We regard this qualified answer as more valuable than an inflated universal one would have been, for a specific, falsifiable reason: this project’s own predecessor codebase claimed graph topology as “the differentiator” on every dataset it touched, and that claim collapsed on inspection into 0.0 feature importance and leakage-inflated metrics once anyone checked. A four-for-four sweep of positive results, produced by the same team that inherited that history, would be the least trustworthy possible outcome of this study, not the most impressive one. The graph-based fraud detection literature this paper positions itself within has, with some exceptions, tended to report uniformly positive results for structural features [1], [2]; a controlled, same-architecture, leakage-audited ablation that instead finds two positives, one null, and one negative result – and says so, including naming which of its own numbers fall inside its own stated noise floor – is offering a more falsifiable, and therefore more useful, claim than a paper that reports success everywhere it looked.

B. A second, independent contribution: production engineering

The paper’s second contribution does not depend on the ablation’s outcome at all: a Procure-to-Pay ERP core with a forensic audit console was designed, built, and *measured*, not merely specified, against the hardening requirements a real deployment would need. The online scoring path reuses the research pipeline’s own WindowGraph and LapMemory classes directly rather than reimplementing them, verified by a parity test that replays 20,000 real transactions through both the batch and online computation paths with zero mismatches across all thirteen topology columns (§III-H1). Financial-integrity invariants – double-entry balance, fiscal-period close, and audit-log immutability – are enforced as Postgres triggers the application layer cannot bypass, not as conventions a future migration could quietly break, and a five-layer duplicate-payment defense was proven, not asserted, under genuine concurrency: 30 simultaneous identical payment requests against one invoice produced exactly one successful payment and 29 correctly rejected duplicates (§III-H2). Performance and failure behavior are reported as measured, including where the measurement is unflattering: the mixed workload ramp clearly degrades before 500 concurrent users on a single dev-mode process (p95 8.18s against a 1s target), and an initial backup/restore drill correctly surfaced a Postgres MVCC-driven fingerprint mismatch under concurrent writes rather than hiding it. None of this is contingent on topology winning its ablation anywhere; it is a claim about engineering discipline – that a leakage-free feature computation can be carried, unmodified, from a research pipeline into a live scoring path, and that the resulting system’s integrity guarantees, failure modes, and performance ceiling can be demonstrated under real load rather than assumed. We consider a genuinely deployed, tested, database-enforced system a different and complementary kind of evidence from a PR-AUC number, and this paper reports both rather than treating the console as an illustrative demo of the research result.

C. Future work

Three directions follow directly from specific, named gaps this build surfaced, rather than from general aspiration. First, and most concretely scoped: persist the TGN’s per-account GRU memory tensor as a checkpointed, incrementally-updatable artifact, so that a live transaction can update and reuse account memory in $O(1)$ rather than requiring a full history replay. This is the single change that would let a true online champion – topology plus a learned temporal representation – be served at all; today it is blocked by an engineering gap (no persisted memory), not by an algorithmic one, and closing it would also let the online/offline gap quantified in Table XVI (0.005–0.208 PR-AUC by dataset) be measured directly rather than only bounded. Second, extend multi-seed evaluation, and ideally formal interval estimation, to the ablation itself rather than only to the champion selection step: the seven-arm decomposition that produces this paper’s headline lift numbers remains single-seed, and work on uncertainty-aware evaluation for graph-based fraud models [7]

points toward replacing this project’s informal ~ 0.01 noise-floor heuristic with a proper bootstrap or paired confidence interval, particularly for SAP Würzburg, where a 46% relative standard deviation in a related multi-seed experiment suggests the ablation’s point estimate alone should not be over-trusted. Third, resolve the batch pipeline’s z-score instability upstream rather than leaving it as a disclosed but uncorrected defect: replacing the two-pass variance formula with Welford’s algorithm – already done in the online engine – would remove the one place in this study where the offline research numbers and the online serving numbers are known to diverge by construction, at the cost of recomputing every headline PR-AUC this paper reports. We view that recomputation, not the current disclosure-only stance, as the right eventual resting state, and name it as such rather than treating disclosure as a substitute for the fix.

The narrower, more durable claim this paper defends is not that graph topology is or is not valuable for fraud detection in general, but that the question can be asked honestly, with leakage structurally excluded by construction rather than checked for after the fact, and that the answer one gets is worth reporting exactly as measured – including when it says no.

REFERENCES

- [1] Z. Yuan *et al.*, “A comprehensive survey on gnn-based anomaly detection,” *ACM Computing Surveys*, vol. 57, no. 3, pp. 1–39, 2025.
- [2] S. Motie and B. Raahemi, “Financial fraud detection using graph neural networks: A systematic review,” *Expert Systems with Applications*, vol. 240, p. 122557, 2024.
- [3] R. R. Devi, J. E. Raja, and Y. B. Chin, “RI-gnn fusion for real-time financial fraud detection,” *Scientific Reports*, vol. 15, no. 1, p. 1042, 2025.
- [4] G. Tong *et al.*, “Financial transaction fraud detector based on imbalance learning and gnn,” *Applied Intelligence*, vol. 53, no. 14, pp. 17820–17835, 2023.
- [5] L. Zhi *et al.*, “Imbalanced fraudulent transaction detection based on embedding-aware conditional gan,” *Journal of Big Data*, vol. 12, no. 1, pp. 1–22, 2025.
- [6] M. Boyapati *et al.*, “BalancerGNN: Balancer GNNs for imbalanced datasets,” *IEEE Access*, vol. 13, pp. 14201–14215, 2025.
- [7] Y. Shi *et al.*, “Uncertainty-aware spatio-temporal contrastive GNNs for financial fraud detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-26)*, vol. 40, 2026, pp. 1420–1428.
- [8] N. Innan *et al.*, “Financial fraud detection using quantum graph neural networks,” *Quantum Machine Intelligence*, vol. 6, no. 1, pp. 1–14, 2024.
- [9] M. Grossi *et al.*, “Mixed quantum-classical method for fraud detection with quantum feature selection,” *IEEE Transactions on Quantum Engineering*, vol. 3, pp. 1–12, 2022.
- [10] H. Wang, N. Liu, and Q. Tan, “Reinforced causal explainer for graph neural networks,” in *Proceedings of the ACM Web Conference 2023 (WWW ’23)*, Austin, TX, USA, 2023, pp. 432–442.
- [11] D. Cheng, Y. Zou, S. Xiang, and C. Jiang, “Graph neural networks for financial fraud detection: A review,” *Frontiers of Computer Science*, 2025, arXiv:2411.05815.
- [12] B. Deprez, W. Wei, W. Verbeke, B. Baesens, K. Mets, and T. Verdonck, “Advances in continual graph learning for anti-money laundering systems: A comprehensive review,” arXiv:2503.24259, 2025.
- [13] Y. Lucas and J. Jurgovsky, “Credit card fraud detection using machine learning: A survey,” arXiv:2010.06479, 2020.
- [14] Y. Chen, C. Zhao, Y. Xu, C. Nie, and Y. Zhang, “Year-over-year developments in financial fraud detection via deep learning: A systematic literature review,” arXiv:2502.00201, 2025.
- [15] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson, “Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics,” arXiv:1908.02591, 2019.
- [16] X. Li, S. Liu, Z. Li, X. Han, C. Shi, B. Hooi, H. Huang, and X. Cheng, “Flowscope: Spotting money laundering based on graphs,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-20)*, vol. 34, no. 4, 2020, pp. 4731–4738.
- [17] H. Tariq and M. Hassani, “Topology-agnostic detection of temporal money laundering flows in billion-scale transactions,” arXiv:2309.13662, 2023.
- [18] A. N. Eddin, J. Bono, D. Aparicio, D. Polido, J. T. Ascensao, P. Bizarro, and P. Ribeiro, “Anti-money laundering alert optimization using machine learning with graphs,” arXiv:2112.07508, 2021.
- [19] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 2, pp. 318–327, 2020, originally presented at ICCV 2017.
- [20] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [21] N. Japkowicz and S. Stephen, “The class imbalance problem: A systematic study,” *Intelligent Data Analysis*, vol. 6, no. 5, pp. 429–449, 2002.
- [22] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [23] H. Wu, R. Han, and Y. Yuan, “Vae-inf: A statistically interpretable generative paradigm for imbalanced classification,” arXiv:2604.25334, 2026.
- [24] E. Rossi *et al.*, “Temporal graph networks for deep learning on dynamic graphs,” in *ICML Workshop on Graph Representation Learning and Beyond*, 2020, pp. 1–12.
- [25] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations (ICLR)*, 2015.
- [26] Y. Wang, Y.-Y. Chang, Y. Liu, J. Leskovec, and P. Li, “Inductive representation learning in temporal networks via causal anonymous walks,” arXiv:2101.05974, 2021.
- [27] M. Dileo, M. Zignani, and S. Gaito, “Durendal: Graph deep learning framework for temporal heterogeneous networks,” arXiv:2310.00336, 2023.
- [28] L. Yu, L. Sun, B. Du, and W. Lv, “Towards better dynamic graph learning: New architecture and unified library,” arXiv:2303.13047, 2023.
- [29] Y. Lei, Q. Shen, and J. Song, “A multi-scale graph learning framework with temporal consistency constraints for financial fraud detection in transaction networks under non-stationary conditions,” *International Journal of Data Science and Analytics*, 2026, arXiv:2603.14592.
- [30] M. A. Vasarhelyi, M. G. Alles, and K. T. Williams, “The acceptance and adoption of continuous auditing by internal auditors,” *International Journal of Accounting Information Systems*, vol. 13, no. 3, pp. 267–281, 2012.
- [31] Z. N. Jawad *et al.*, “ML-driven optimization of ERP systems: A comprehensive review,” *Discover Artificial Intelligence*, vol. 4, no. 1, pp. 1–28, 2024.
- [32] X. J. Mamakou *et al.*, “Post-implementation evaluation of ERP systems: An internal auditors’ perspective,” *Information Systems Management*, vol. 41, no. 2, pp. 180–198, 2024.
- [33] J. Tritscher, F. Gwinner, D. Schloer, A. Krause, and A. Hotho, “Open ERP system data for occupational fraud detection,” arXiv:2206.04460, 2022.
- [34] M. Schreyer, T. Sattarov, C. Schulze, B. Reimer, and D. Borth, “Detection of accounting anomalies in the latent space using adversarial autoencoder neural networks,” arXiv:1908.00734, 2019.
- [35] H. Hemati, M. Schreyer, and D. Borth, “Continual learning for unsupervised anomaly detection in continuous auditing of financial accounting data,” arXiv:2112.13215, 2021.
- [36] P. L. Foale, L. Da Silva, F. Khomh, H. Li, and E. Merlo, “Logging requirement for continuous auditing of responsible machine learning-based applications,” arXiv:2508.17851, 2025.
- [37] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 2017, pp. 4765–4774.
- [38] S. M. Lundberg *et al.*, “From local explanations to global understanding with explainable AI for trees,” *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.
- [39] N. H. Dao, “Explainable fraud detection with GNNExplainer and SHapley values,” arXiv:2509.12262, 2025.
- [40] F. Almalki and M. Masud, “Financial fraud detection using explainable AI and stacking ensemble methods,” arXiv:2505.10050, 2025.

- [41] A. Faccia, "Explainable ai (xai) in auditing: Bridging the gap between predictive fraud models and regulatory standards," *Journal of Risk and Financial Management*, vol. 19, no. 5, p. 311, 2026.
- [42] M. N. Uddin and M. M. Aziz, "Shapley value-guided adaptive ensemble learning for explainable financial fraud detection with u.s. regulatory compliance validation," arXiv:2604.14231, 2026.
- [43] H. Ye, A. Laxman, Y. Yuan, K. Flautner, and N. Talati, "Blazingaml: High-throughput anti-money laundering (aml) via multi-stage graph mining," arXiv:2604.12241, 2026.
- [44] A. Premkumar *et al.*, "Leveraging artificial intelligence for enhanced anti-money laundering in financial institutions," *Discover Computing*, vol. 29, p. 240, 2026.
- [45] Y. Tian, G. Liu, J. Wang, and M. Zhou, "Transaction fraud detection via an adaptive graph neural network," arXiv:2307.05633, 2023.
- [46] H. R. Roth *et al.*, "Privacy-preserving federated fraud detection in payment transactions with nvidia flare," arXiv:2603.13617, 2026.
- [47] M. S. Uddin, M. H. Amin, N. J. Ema, B. Uddin, T. Ahmed, and A. H. Zidan, "Multilingual financial fraud detection using machine learning and transformer models: A bangla-english study," arXiv:2603.11358, 2026.
- [48] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 2017, pp. 5998–6008.
- [49] C. Wang, C. Nie, and Y. Liu, "Evaluating supervised learning models for fraud detection: A comparative study of classical and deep architectures on imbalanced transaction data," arXiv:2505.22521, 2025.
- [50] S. Wang, T. Tricco, X. Jiang, C. Robertson, and J. Hawkin, "Synthetic demographic data generation for card fraud detection using gans," arXiv:2306.17109, 2023.
- [51] M. Zhu, Y. Gong, Y. Xiang, H. Yu, and S. Huo, "Utilizing gans for fraud detection: Model training with synthetic transaction data," arXiv:2402.09830, 2024.
- [52] A. F. Ghahfarokhi, T. Mansouri, M. R. S. Moghadam, N. Bahrambeik, R. Yavari, and M. F. Sani, "Credit card fraud detection using asexual reproduction optimization," arXiv:2306.01008, 2023.
- [53] C. Giusti, L. Guarnera, M. Casu, and S. Battiato, "Fraud is not just rarity: A causal prototype attention approach to realistic synthetic oversampling," arXiv:2507.14706, 2025.
- [54] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY: Springer-Verlag, 2006.
- [55] S. Jesus *et al.*, "Turning the tables: Biased, imbalanced, dynamic tabular datasets for ml evaluation," in *Advances in Neural Information Processing Systems (NeurIPS 2022)*, 2022.
- [56] E. Altman *et al.*, "Realistic synthetic financial transactions for anti-money laundering models," *Advances in Neural Information Processing Systems (NeurIPS Datasets and Benchmarks)*, 2023.
- [57] E. A. Lopez-Rojas and S. Axelsson, "Banksim: A bank payment simulation for fraud detection research," in *Proceedings of the 26th European Modeling and Simulation Symposium (EMSS)*, 2014.
- [58] E. A. Lopez-Rojas, A. Elmir, and S. Axelsson, "Paysim: A financial mobile money simulator for fraud detection," in *Proceedings of the 28th European Modeling and Simulation Symposium (EMSS)*, 2016.
- [59] A. Salih, S. T. Zeebaree, S. Ameen, A. Alkhyat, and H. M. Shukur, "A survey on the role of artificial intelligence, machine learning and deep learning for cybersecurity attack detection," in *2021 7th International Engineering Conference (IEC)*. IEEE, 2021, pp. 61–66.
- [60] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [61] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 2017, pp. 3146–3154.
- [62] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [63] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations (ICLR)*, 2017.
- [64] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.
- [65] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 2017, pp. 1024–1034.
- [66] W. Hamilton, *Graph Representation Learning*. San Rafael, CA: Morgan & Claypool Publishers, 2020.
- [67] F. Provost, T. Fawcett, and R. Kohavi, "The case against accuracy estimation for comparing induction algorithms," in *Proceedings of the 15th International Conference on Machine Learning (ICML)*, 1998, pp. 445–453.
- [68] J. Davis and M. Goadrich, "The relationship between precision-recall and roc curves," in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006, pp. 233–240.
- [69] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [70] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," in *Advances in Neural Information Processing Systems (NeurIPS 2014)*, 2014, pp. 2672–2680.
- [71] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *International Conference on Learning Representations (ICLR)*, 2014.
- [72] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.
- [73] P. Helland, "Idempotence is not a medical condition," *ACM Queue*, vol. 10, no. 4, pp. 30–46, 2012.